

Beyond the Baseline: How the Use of Machine Learning Models Enhances Win Prediction in NBA Basketball

A dissertation submitted in partial fulfilment of the requirements for the degree of Master of Science.

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Declaration

I hereby declare that this dissertation entitled 'Beyond the Baseline: How the Use of Machine Learning Models Enhances Win Prediction in NBA Basketball' is entirely my own work and it has never been submitted nor is it currently being submitted for any other degree.

Abstract

In recent years, the predictive modelling in professional basketball has evolved from simple statistical estimation to advanced machine learning (ML) applications capable of capturing the strategic complexity of the sport. This study introduces a data-driven framework for NBA win prediction using classifiers trained on team-versus-opponent performance differentials. The objective is to assess whether ML models can outperform the Pythagorean Expectation (Pe) approach under varying competitive conditions.

Several supervised algorithms were deployed; including logistic regression, random forest, XGBoost, neural networks, and ensemble hybrids across three settings: regular season, postseason, and simulated playoff-calibre contests. All model inputs were structured around engineered features capturing possession control, shooting efficiency, and rebounding balance.

Empirical results show that adaptive classifiers consistently surpass traditional Pe models in high-intensity playoff simulations where tactical volatility increases. Defensive rebounding, effective field goal percentage, turnover differential, and three-point accuracy were identified as dominant predictors of win probability. Ensemble and neural network approaches demonstrated superior performance, with gains reaching up to 10% accuracy over Pe benchmarks. Limitations around team-level granularity, potential overfitting, and model transparency are acknowledged.

Recommendations include extending the framework to include player-level KPIs, incorporating graph-based representations of team matchups, and applying interpretability tools from Explainable AI. Overall, this study enhances the methodological selection in basketball analytics, supporting richer tactical diagnostics and more accurate forecasting in competitive sports environments.

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Table of Contents

Declaration	ii
Abstract	iii
Acknowledgements	iv
The List of Tables	vi
The List of Figures	vii
Chapter 1.0 – Introduction.....	1
Chapter 2.0 – Literature Review	2
2.1 - Machine Learning & AI Applications	6
Chapter 3.0 – Methodology.....	10
3.1 - Data Collection.....	11
3.2 - Pre-Processing & Feature Engineering	11
3.3 - Pe Model Development	12
3.4 - ML Model Implementation.....	13
3.5 - Evaluation & Visualisation	16
Chapter 4.0 - Results	16
4.1 - Performing Pe	18
4.2 - ML Applications	22
Chapter 5.0 – Discussion	26
5.1 – Interpretations of Key Findings & Comparison to Literature	26
5.2 – Theoretical & Conceptual Linkages	28
5.3 – Research Limitations & Reasonings	29
5.4 – Implications for Practice.....	31
5.5 – Link to Research Questions and Future Directions.....	32
Chapter 6.0 – Conclusion	33
Glossary & Table of Abbreviations	34
Appendix A – Pe Fitted vs Actual Win% in 2022-2025 (Regular Season)	38
Appendix B – Pe Weibull vs Actual Win% in 2022-2025 (Regular Season)	39
Appendix C – Pe Fitted vs Actual Win% in 2022-2025 (Postseason)	40
Appendix D – Pe Weibull vs Actual Win% in 2022-2025 (Postseason)	41
Appendix E – Research Project Ethics Form Approval	42
Appendix F – Github Repository Link	50
Bibliography	51

The List of Tables

Table 1: *Game Counts for Each Classification of Game Type during the Regular Season.*

Table 2: *Functional Overview of ML Models Within the Analytical Framework.*

Table 3: *Seasonal exponent estimates using Fitted & Weibull derived Pe models in the regular and post season contexts.*

Table 4: *Validation of Pythagorean Expectation Model: Predicted vs Actual Win Percentages Across Seasons.*

Table 5: *Contextual Model Validation: Comparative Pe Performance Across Matchup Types.*

Table 6: *Features initially selected through the feature importance process.*

Table 7: *Aggregated Feature Ranking Across Machine Learning Models: Contextual Stat Importance in Predicting Win Outcomes.*

Table 8: *Aggregated Feature Ranking Across Machine Learning Models: Contextual Stat Importance in Predicting Win Outcomes in ‘Playoff vs Playoff’ context.*

Table 9: *Regular Season Model Performance Metrics (2025 Validation).*

Table 10: *Post Season Model Performance Metrics (2025 Validation).*

Table 11: *Playoff Vs Playoff Model Performance Metrics (2025 Validation).*

The List of Figures

Figure 1: *The formula provided by Kaplan & Rich, (2017) for detailing the exponent for the Pythagorean expectation equation.*

Figure 2: *Expression of the probability density of the three-parameter Weibull distribution, detailed in Almeida et al., (2024).*

Figure 3: *Estimated Pythagorean Exponents by Season (2022–2025): Fitted vs Weibull-Derived Pe Models.*

Figure 4: *Pythagorean Expectation Formula for Team Win Percentage Estimation.*

Figure 5: *Postseason Pe Model Residuals: Actual Win% vs Predicted Win% in 2023 & 2024.*

Figure 6: *Postseason Win% vs Point Differential: Pe Model Comparison Across Seasons 2023 & 2024.*

Figure 7: *Permutation Test Distribution of R^2 Scores: Validating 'PE_Weibull' Feature Impact (2025 Regular Season).*

Figure 8: *Correlation Matrix of Context-Adjusted Performance Metrics: Evaluating Feature Relationships and Relevance to Team Wins.*

Chapter 1.0 – Introduction

In professional sports, being able to accurately understand where winning is driven from is a powerful tool for teams and organisations to be able to deploy. In the most competitive environment of the National Basketball Association (NBA), gaining an edge can translate into extensive benefits. Traditional approaches such as the Pythagorean Expectation (Pe) have provided interpretable baselines rooted in point differentials and have been tested in prior literature ascertaining to be more than capable to model outcomes effectively. However, as stated by the study of Baumer et al., (2023), the Pe model can lack depth in its analytic toolkit and this study focuses these shortcomings implementing a machine learning (ML) framework that evaluates performance through differentials across box score and advanced metrics.

The predictive system integrates a range of ML algorithms that includes logistic regression (LR), random forest (RF), XGBoost, feedforward neural networks (FNN), and hybrid ensemble techniques. These models support vector machines and deep learning methods that are progressively utilised to calculate game outcomes and team performance (Bunker & Susnjak, 2022). The models were selected for their strengths in learning from game-level data that is not always intrinsically linked to the direct outcome. Model evaluation is performed across three competitive domains: regular season, postseason, and simulated “Playoff vs Playoff” (PvP) matchups, providing a fresh and unique view of predictive outcomes. By transforming raw statistics into team-versus-opponent differentials, and pruning redundant or signal-leaking features, this study ensures predictive robustness. This is encouraged by the study of Nguyen et al., (2022), who looked at similar methods of crafting differentials to creating more meaningful distinctions between predictive signals and redundant features.

The central objective of the study is to extract and compare the feature importance rankings across models and contexts, revealing which statistical fields signal consistent win outcomes. Together, these modelling strategies aim to establish a reproducible pipeline that influences performance metrics, playoff-aware segmentation, and comparative model evaluation to improve both predictive power and decision-making clarity for professional basketball practitioners. The study offers a flexible framework for forecasting win probability while maintaining methodological transparency through calibrated baselines and explainable feature logic, as well as establishing that

ML capabilities can enhance the baseline set by Pe. Furthermore, the study looks to forge its own path of analysis, while not disregarding what has been discovered previously in this area of study, such as the works of (Cai et al., 2019), (Bunker and Thabtah, 2019) and (Osken and Onay, 2022).

The formation of the research study is as follows: chapter 1 introduces the research aims and rationale. Chapter 2 surveys key literature surrounding basketball analytics, the Pe model and ML applications, with emphasis on feature engineering, model calibration, and predictive strategies. Chapter 3 presents the dataset, preprocessing pipeline, and feature generation techniques used to construct differential performance indicators across matchup types, with chapter 4 outlining the results of the Pe and ML models compared across a variety of contexts, alongside analysis of feature importance rankings and outcome-driven insights. Chapter 5 is a critical discussion of the findings, evaluating all model performances, feature dynamics and contextual behaviours to assess their practical implications. Finally, chapter 6 concludes the study by summarising key findings, highlighting methodological contributions, and discuss future directions for predictive basketball modelling.

Chapter 2.0 – Literature Review

Traditional models for analysing basketball statistics have employed a variety of methods through the last four decades. The Pe has long served as an intuitive tool to aggregate a teams win percentage; the tool is based on the total of points scored and points allowed by a given team. Initially developed by Bill James in the 1980s for baseball, the model has been adapted for many other sports, including basketball since its inception. As stated by Oliver, (2003) the core adaptation required for the model to be functional is a sport-specific adjustment to be made in the identification and implementation of the exponent portion of the equation. These traditional approaches provide greater interpretability and simplistic baseline for team and player performance evaluations to be made and visualised.

This understanding has evolved over time, such as the study of Caro and Machtmes, (2013) and Ehrlich et al., (2021), which theoretically justify and empirically calibrate the base model's parameters in efforts to enhance the model's predictive capabilities. Despite being based around different sports, these studies are applicable when considering how best to develop any Pe model for predictive accuracy. By situating their analyses within distinct sporting environments; NCAA

American football and Major League Baseball respectively, both studies demonstrate that the foundational structure of the Pe model needs to be critically re-evaluated and adapted accordingly for the model's deployment in each individual competitive setting.

Building on foundational efforts to calibrate and evaluate the Pe model across sports domains, Boudreaux et al., (2021) advanced this effort by testing multiple contest success functions (CSFs) against MLB team data, demonstrating that model calibration has a substantial effect on predictive accuracy. These findings underscore a methodological shift toward identifying functional forms that better reflect the underlying scoring dynamics of a sport. This perspective lays a foundation for studies to explore how exponent behaviour in Pe models is shaped by sport-specific scoring structures, a theme that is central to the approach taken by Kaplan & Rich, (2017). Who details a theoretical understanding of decomposing the Pe model by deriving a singular team-specific win expectation equation. This proves that the key exponent, typically expressed as (γ), is reliant on the average points per game (R_{total}) and the typical winning margin each individual team has. This expression is detailed in Figure 1, where (β) is the regression slope from a linear approximation of win percentage against the points differential.

Figure 1: The formula provided by Kaplan & Rich, (2017) for detailing the exponent for the Pythagorean expectation equation.

$$\hat{\gamma} = 4 \times R_{total} \times \hat{\beta},$$

This formula rationalises why the exponent is far greater in basketball than in baseball, showing the studies analytical approach highlights the exponent is not an arbitrary constant. Rather the study details that the exponent is a parameter reflecting sport-specific scoring dynamics that sets the mathematical foundation for parameter calibration, a critical component for a valid model to be constructed.

Building on this theoretical framework, Yoon and Choi, (2022) compare deterministic and statistical approaches to establishing the Pythagorean exponent. The study shows that regression-based estimation yields an exponent that is slightly lower than its deterministic counterpart, with a larger correlation between the regression coefficients of points scored and points allowed. This showed that the improved predictive accuracy can be accounted to fluctuations in game outcomes over the

course of a full season. Furthermore, this perspective captures the inherent variability in scoring profiles of each team. The studies correlation analysis establishes a strong relationship between scoring differentials and the calculated exponent, reinforcing the necessity of adaptive modelling strategies that dynamically adjust predictive parameters based on every-changing game conditions. This extends to all iterations of the Pe model across sporting domains, with studies by Vine, (2016) in cricket and Baysal and Yildiztepe, (2019) in European football serving as clear examples of its ability to be adaptable to the scoring natures of differing sports.

Chen and Li, (2016) further advance the discourse by investigating the phenomenon of shrinkage in the Pythagorean exponent in a predictive setting. Their findings illustrate that early-season evaluations often overestimate team strength, leading to inflated win expectations that gradually stabilise as the randomness of game results diminishes throughout the length of a full season. The study states that “the shrinkage factor diverges from approximately 0.5 in the early season to practically 1.0 by the season’s end”. Thus, proving the necessity of adapting predictive models consistently and that the study stresses that predictive precision tends to decline as the season progresses, which is counterintuitive to natural assumptions that additional data improves prediction. This effect is attributed to possible changes in team dynamics, player fatigue, injuries, and strategic shifts among playoff-contending teams, which can disrupt predictive stability.

The study also finds that when comparing different metrics, Point Ratio (PtR) and Point Difference (PtD) consistently outperformed Win-Loss Ratio (WLR) when predicting results. This insight reinforces the argument that predictive models should prioritise scoring-based variables, rather than win-loss metrics. Their conclusions support the notion that constant adjusting of the exponent throughout the season, as well as from season to season, can substantially improve live and mid-season predictive capabilities. This is complemented by the work of Michailidis et al., (2025) that explored the application of the Pe in Euroleague basketball. Their study identified an exponent value of 11.19, which is significantly smaller than the estimations made for the NBA. By employing bootstrap methods, they generate confidence intervals that ranged from 9.98 to 12.32, which quantified the uncertainty in the estimated exponent. This robust statistical approach strengthens the notion of needing to recalibrate the model for the differing scoring characteristics that can occur in different points of the season and emphasises the importance of adapting analytic tools to the unique features of different leagues.

Almeida et al., (2024) expands on the traditional model by modelling points scored and allowed as independent random variables drawn from three-parameter Weibull distributions, each with its own distinct shape parameter. The Weibull distribution is a constant probability distribution that is frequently used to model time-to-failure data and other phenomena where the rate of occurrence may change over time (Wang et al., 2022a). It is defined by its probability density function, shown in Figure 2.

Figure 2: Expression of the probability density of the three-parameter Weibull distribution, detailed in Almeida et al., (2024).

$$f(x; \alpha, \beta, \gamma) := \frac{\gamma}{\alpha} ((x - \beta)/\alpha)^{\gamma-1} e^{-((x-\beta)/\alpha)^\gamma}$$

The equation states that (α) is the scale parameter, (β) is the location parameter and (γ) is the shape parameter that governs the form of the distribution. The allow the shape parameter to differ for runs scored and runs allowed, the study captured a nuanced picture of scoring dynamics compared to models that assume identical distributions for both. Their approach employs the Method of Moments to numerically estimate these parameters, resulting in improved predictive capabilities. While the study is grounded in baseball, its computation techniques and calibration methodology can offer a blueprint for adapting the studies model to a basketball dataset where differences in game tempo, scoring variability, and defensive strategies require precise parameter tuning. These advancements would have a similar impact on optimising a model to evaluate player valuation, roster composition and overall predictive competencies.

Sarlis & Tjortjis, (2020) take further steps towards understanding how to improve predictive competencies that can extend beyond the abilities of a Pe led system. They offered a comprehensive review of common box-score statistics and advanced metrics, such as Plus-Minus, Player Impact Estimate (PIE) and Wins Above Replacement (WAR). The study argues the advantages of a multidimensional approach to evaluating performance is needed to capture the complexities that are observed in a hyper-dynamic sporting environments, such as the NBA. An emphasis is placed on player synergy, matchup dependencies and defensive contributions to underscore that no single metric suffices to give a full perception of a team or players performance. Rather that combining

these metrics minimises bias that can arise from isolation statistics and can provide a more complete narrative to understand where performance originates from.

For many of these studies, the inherent benefit is that traditional methods of statistical analysis provide clear and reliable results that come from high levels of interpretability in relation to showcasing relevant insights to decision-makers, such as coaches or owners of professional sports teams. However, the studies detailed in this section also shows that there are limitations to the approach of Pe. The methods struggle when handling the dynamic outcomes displayed by season-long data. As stated by Michailidis et al., (2025), the model works for each team independently and rather focuses on how each team works intrinsically than enforcing a constraint to evaluate how they perform against each other. Modern ML tools can offer compelling avenues to enhance these established models. By blending the interpretability of traditional models with the predictive power of ML tools, researchers can achieve greater accuracy and provide more actionable insights for decision-makers to act upon.

2.1 - Machine Learning & AI Applications

The rapid evolution of Artificial Intelligence (AI) and ML have profoundly reshaped the landscape of sports analytics, moving beyond rudimentary statistical analysis to sophisticated predictive models (Wang, 2023). The multi-layered and dynamic nature of modern team sports which is influenced by a myriad of complex factors, often eludes the grasp of traditional methodologies. The imperative for more adaptable and robust predictive tools has therefore propelled the integration of ML algorithms, capable of discerning non-linear patterns and intricate interactions within high-dimensional datasets. A diverse array of ML algorithms has been deployed in this pursuit, yielding varied accuracies. ML methods such as Logistic Regression (LR), Decision Trees (DT), and Random Forest (RF) are widely applied across various team sports due to their versatility in handling complex datasets.

The role of feature selection is a vital first step to informing any ML algorithm. Horvat and Job, (2019) emphasises the minimalistic nature of naive classification models, relying primarily on win-loss ratios extracted via structured historical segmentation. The study's findings suggest that features when grounded in recency and context can yield predictive accuracy, particularly when training windows are calibrated intentionally. In contrast, the study by Plakias et al., (2024) adopts a more

robust ML framework, incorporating dimensional reduction and interpretability techniques. This methodological depth enables the identification of key performance indicators (KPIs) that dominate under differing competitive pressures. The juxtaposition of these approaches underscores the evolving landscape of feature engineering in sport analytics. This is comparable to the study of Topcu & Çilengiroğlu, (2024), which examined the performance of a variety of predictive algorithms that included LR, variations of DT and RF models on a vast dataset. By incorporating KPI's the researchers critically evaluated how a variation of KPI's affected predictive performance. The study found variations of power metrics, achieving around 62-75% accuracy. While based on football statistics, the methods that the study uses are relevant throughout team sports. The successful integration of DT and RF highlights the advantages of using structured models alongside traditional statistical methods as an approach that enhances predictive modelling in team sports.

Similarly, Andrews et al., (2021) explores match predictive modelling using ML libraries LR, Support Vector Machines (SVM) and XGBoost on an extensive Premier League football dataset. Their findings are that LR can perform competitively with the advanced ensemble methods when comparing predictive accuracy and computational load. This study demonstrates that the use of statistically grounded models alongside complex algorithms can improve the reliability of the results. This indicates a pattern of best practices, that leveraging a mixture of structured approaches with robust ensemble techniques can refine outcome predictions and contribute actionable insights.

The survey studies of Sukumaran et al., (2022) and Gálvez et al., (2024) showcased a variety of different publications that had a variety of results that came from using differing ML methods. The review highlights the application of LR and Hybrid Fuzzy Support Vector Machines (HFSVM) in the study of Kaur & Jain, (2017), where they found that with robust feature selection that the HFSVM model returned an 88.26% predictive accuracy. This work emphasises the need for integrating multiple performance metrics and optimising feature selection to enhance predictive performance.

In a complementary vein, Agbozo et al., (2024) offers a thorough overview that catalogues current case studies that employed AI applications in basketball. They highlight that future research could focus on integrating explainable AI technologies to boost model understanding, while exploring multi-modal data fusion that combines a variety of analytical functions to enrich situational awareness. The advantages of constructing these adaptive models in this manner can capture

dynamic changes in team strategy, through the evaluation of areas that can construct a more nuanced approach to team chemistry implications.

Ke et al., (2024) proposes an innovative approach that merges unsupervised and supervised learning techniques to support roster optimisation strategies in the NBA and WNBA. This is achieved by applying principal component analysis (PCA) to decrease dimensionality in player performance to be used in clustering methods to organise players into distinct categories. This clustering highlights differences in playing styles and positions, as well as distinguishing players into 'level' clusters for the purpose of identifying the players that fall into the 'elite' cluster. In the subsequent supervised phase, a neural network model is then employed to predict player and thus team performance. Feature selection is vital to this process; by intuitively reducing features to only the most necessary KPI's that drive team success, the model gains interpretability, computational efficiency and improved accuracy in assembling a cost-effective, yet performance driven roster. This links heavily with the study of Thabtah et al., (2019), as both studies highlight the critical importance of rigorous feature selection prior to integration of ML tools.

Thabtah's study develops an outcome prediction model for NBA games using classifiers Naïve Bayes, Artificial Neural Network (ANN) and DT to establish the truly knowing statistical categories that influence the outcomes of games. This process involves the selection of key variables, pushed through two different filtering methods and a rule induction algorithm. These are identified as Correlation Feature Set (CFS), Multiple Regression and the Repeated Incremental Pruning to Produce Error Reduction (RIPPER) algorithm. By employing these three distinguishing approaches, the utmost vital features can be acknowledged. These conditions allowed for the identification of the most vital factors, which were defensive rebounds (DRB), 3-point percentage (3P%) and total rebounds (TRB), that improved game outcome forecasts by around 2-4%.

This aligns well with the study of Ke, where PCA and clustering can place emphasis on reducing noise and irrelevant features. Combining these studies illustrates that combining robust feature selection heavily affects predictive capabilities and thus allowing teams to make informed decisions about roster construction. The synergy of these studies identifies that a unified approach to player and team evaluation ensures that predictive models capture both immediate, single game impacts as

well as long-term roster implications, ultimately allowing coaches and team managers to make better informed decisions.

Chen et al., (2025) reviews the prevailing reliance on static, aggregate-level data in basketball research, highlighting underutilisation of event and tracking data despite their capacity to capture dynamic game flow. The studies systematic review accentuates a critical need for feature engineering and temporal sensitivity, issues directly addressed in the rule-based study by Iatropoulos et al., (2025), who apply association rule mining to 20 seasons of NBA data to uncover KPI patterns that influence game outcomes. The study's findings reveal temporal variability in KPI impact, with heightened defensive efforts in fourth quarter and overtime scenarios. Complementing this, Liang et al., (2025) provides a comparative analysis of NCAA and NBA games using classification models to determine discriminant KPIs, demonstrating that metrics like TS%, PER, and DRTg exhibit strong predictive power across all game states. Collectively, these studies advocate for a performance modelling paradigm rooted in KPI granularity and principles that underpin ML pipelines aimed at forecasting team success.

Papageorgiou et al. (2024) focuses on forecasting individual player performance using a widespread evaluation of 14 ML models across 18 advanced KPI's. The study is based on a cleaned pool of 90 high-performance NBA players, employing game-lag feature engineering and proposes a tailored ranking metric called weighted average percentage error (WAPE). The study reveals that tree-based models, in particular randomized trees such as Extra Trees (ET) consistently outperformed linear-based approaches in predicting metrics like points, rebounds, assists, and derived KPIs such as Fantasy Points. The studies systematic evaluation underscores the importance of selecting the right ML algorithm, while also taking the critical step of effective feature extraction and transformation, which underpins improved forecasting accuracy and offers actionable insights for decision-makers.

Lampis et al., (2023) details a wide-ranging study on prediction across four major European tournaments. The study leverages a dataset that spans five seasons (2013–2018) and incorporates predictors indicators and rating systems, which include ELO, PageRank, and pi-rating. By comparing ML methods such as LR, RF, and extreme gradient boosting with a traditional baseline model, the study demonstrates that carefully tuned features can improve prediction accuracy by 3–5%. The study emphasises that even in a domain where traditional box-score statistics have been used as

proxies for team performance, the integration of dynamic rating systems can lead to a more balanced and robust predictive model. In a related disposition, Zhao et al., (2023) take a RF algorithm coupled with a graph convolution network (GCN) to predict outcomes of games. Rather than relying solely on vector inputs, that ignore inter-team interactions, the studies approach transforms structured game data into an unstructured graph representation wherein each node represents a team, and edges encapsulate recent match interactions. The spatial structure inherent in the NBA's teams, the studies fused GCN+RF model achieves an accuracy of 71.54%, which was a notable improvement over the baseline GCN model.

Together, these studies illustrate a multi-faceted approach to basketball analytics with Lampis providing a broad framework for basketball outcome prediction through carefully calibrated rating systems and feature engineering, with Zhao pushing the model further by integrating graph-based representations to explicitly account for team interactions and spatial structure. The conjunction of these studies demonstrates that a unified strategy which combines rigorous feature selection, the representation of complex inter-team relationships, and extensive model benchmarking can significantly augment our understanding and prediction capabilities.

Chapter 3.0 – Methodology

This study adopts a pragmatic, analytically grounded approach to develop and validate a predictive framework for modelling NBA game outcomes. Following the conclusion of the 2024–25 season, both historical and near-real-time data are now fully available, providing a robust foundation for model training, analysis, and validation. Recognizing the fundamental differences between regular season and playoff games, this study treats them as distinct yet interrelated domains of analysis. This distinction supports a more inclusive evaluation of predictive performance and informs the design of a framework responsive to tactical shifts and temporal dynamics observed across the highest level of competition.

The inclusion of regular season and playoff games is essential for understanding the full scope of predictive power and contextual sensitivity. As noted in the survey paper by Huyghe et al., (2021), the regular season and postseason differ considerably in statistical and strategic emphasis. Playoff contests are typically shaped by increased defensive intensity and rebounding dominance, whereas regular season outcomes hinge more on offensive efficiency. Mandic et al., (2019) observed that

when playoff data is isolated, key statistical differentials diminish, which corroborates the findings of Huyghe. Further evidence supports this, Ouyang et al., (2024) developed a real-time NBA game prediction model using XGBoost and SHAP-based interpretability techniques. Their analysis of game-level data revealed that predictive accuracy varies meaningfully depending on game phase. SHAP values highlighted dynamic shifts in feature importance depending on whether a game was in its early stages or nearing its conclusion. This notion can be extended across an entire season, reinforcing the argument that playoff games must be treated not as extensions of regular season play but as statistically distinct environments. This meant any predictive framework must be sensitive to the evolving tactical demands and feature distributions specific to each competitive context.

3.1 - Data Collection

Data for this study was accumulated from a dependable, publicly accessible statistical database Basketball Reference, which is the data outlet by SportRadar, “the official stats partner of the NBA, NHL and MLB” (BasketballReference, 2025).

A custom Python-based scraping pipeline was developed to extract relevant data, including box-score and advanced statistical game logs, player metrics and season metadata across both all formats. This scraping process was aimed to emphasis the preservation of game-level granularity and capture basic and advanced performance indicators. Across NBA seasons 2021–22 through 2024–25, a total of 10,533 team-game entries were successfully scraped and compiled into a unified dataset. Each row representing an individual team's statistical perspective on a given matchup. Regular season data accounts for 9,840 entries, while playoff coverage comprises 674 entries, which accounts for 6.4% of total rows in the dataset.

3.2 - Pre-Processing & Feature Engineering

To support robust model development and outcome prediction, the cleaned dataset underwent systematic preprocessing and context-aware feature engineering. Initial preparation involved resolving data inconsistencies, handling missing values, and applying normalization techniques to ensure statistical coherence across all records. Each entry was structurally verified using a team-opponent-date composite key, enabling accurate matchup traceability and preventing duplication. All game-level entries were enriched with opponent-derived statistics such as ‘DRB_opp’, ‘AST_opp’, and ‘TOV_opp’, recalculated using mirrored matchup logic to ensure directional accuracy. These

metrics capture the elaborate tones of team dynamics, which is accentuated through the investigation by Mahmood et al., (2021) and Li et al., (2021). Team branding elements, including standardized abbreviations and logo links, were programmatically integrated to aid downstream visualisation efforts.

Once the cleaning process was completed, the limited size of playoff data was addressed. The study implemented a targeted categorisation system within regular season entries to enrich comparative analysis. This involved identifying whether teams were playoff-qualified or non-qualified in each season and tagging each regular season game accordingly, this grew the available playoff vs playoff matchups exponentially, shown in Table 1.

Table 1: Game Counts for Each Classification of Game Type during the Regular Season.

Matchup Type	Game Count
Playoff Vs Non-Playoff	5124
Playoff Vs Playoff	3360
Non-Playoff Vs Non-Playoff	2030

By distinguishing matchups based on opponent type, the dataset enabled context-aware splits that supported more nuanced pattern detection and volatility analysis. This segmentation extended the effective scope of postseason-like analysis beyond the limited playoff sample, increasing coverage by approximately 398%.

All feature engineering steps were integrated into a reproducible pipeline and version-controlled to maintain methodological transparency. These engineered indicators serve as foundational elements for downstream modelling and visualisation, providing interpretable signals aligned with competitive dynamics.

3.3 - Pe Model Development

As a foundational comparator for the ML framework, the study implemented a series of baseline models focused on the Pe equation. These models were used to establish an interpretable benchmark capable of estimating team win percentages based on scoring dynamics, and to evaluate improvements offered by advanced ML architectures.

To calibrate the Pe exponent, team-level aggregates were generated for each season using total points scored, total points allowed, and binary win labels. The optimal PE exponent was then derived independently for each season using least-squares curve fitting. This multi-season calibration allowed scoring volatility and exponent behaviour to be tracked over time, supporting dynamic benchmarking and contextual sensitivity analysis.

Complementing the fitted exponents, a distribution-based estimation method was applied using a Weibull approach informed by the methodology of Almeida et al., (2024). This technique involved fitting shape parameters to team scoring distributions with 'scipy.stats.weibull_min', where the derived exponent was approximately twice the mean shape parameter 'k'. The Weibull variant provided an alternative interpretation of team strength based on statistical distribution curves, allowing for comparative evaluation with the curve-fitted baseline.

To visually assess the alignment between predicted and actual team outcomes, logo-based scatter plots were constructed for each season. These plots compared fitted Pe estimates with actual win percentages, offering intuitive diagnostics for outlier detection, calibration accuracy, and over- or under-performance analysis.

Both fitted and Weibull-derived exponents were retained as context-aware features and integrated into subsequent ML pipelines. This allowed statistical anchoring of each model's predictions and enabled comparative assessment of interpretable versus distribution-informed logic across competitive environments.

3.4 - ML Model Implementation

Building upon the interpretability and benchmarking capabilities of the Pe models, the study incorporated a collection of supervised ML algorithms to enhance predictive precision and contextual adaptability, aligning with the thoughts of Ahmed et al., (2023). These models were selected for their ability to handle structured game-specific data, accommodate non-linear relationships, and reveal feature-driven variability across distinct competitive phases. These models are ranging from tree-based classifiers to neural networks, which allowed for a comparative evaluation of modelling strategies, with each contributing unique strengths in terms of scalability, pattern learning, and interpretability (see in Table 2).

Table 2: Functional Overview of ML Models Within the Analytical Framework.

Model	Purpose in Framework
Logistic Regression (LR)	Baseline linear classifier used to gauge performance of interpretable statistical logic.
Random Forest (RF)	Robust tree-based model for non-linear classification and early feature salience testing.
Naïve Bayes (NB)	Lightweight probabilistic benchmark to evaluate assumptions of feature independence.
XGBoost	Gradient-boosted algorithm used for precise modelling and SHAP-driven feature analysis.
Feedforward Neural Network (FNN)	Captures abstract representations of team performance through layered feature engineering.
Ensemble (RF + Fuzzy SVM)	Combines multiple classifiers to balance interpretability, precision, and adaptability.

LR was implemented as a linear baseline model to establish predictive power of core features within a straightforward statistical framework. By modelling the relationship between input variables and binary win outcomes, LR offers transparency in decision boundaries and allows for quick calibration of coefficients tied to team-level indicators. Its inclusion ensures that all subsequent models can be benchmarked against a well-understood reference, that is robust in his predictive capabilities.

The RF model served as a non-linear model, chosen for its robustness against overfitting. Using multi-round RF-based screening, the model splits feature space into decision trees, allowing for ensemble-based predictions that remain stable across diverse competitive scenarios. Hyperparameter tuning was conducted using grid search to optimize tree depth, split thresholds, and node constraints, ensuring the model remained scalable while retaining interpretability. The RF classifier provided early insight into feature salience, guiding subsequent ensemble design and hybrid logic integration aligning with the study by Chandru et al., (2025).

NB provided a means to assess the predictive limits beneath the statement of feature independence. Its inclusion offered reference points for model calibration. Even with its simplicity,

NB facilitated fast experimentation and contributed to sensitivity analysis in early modelling iterations.

HFSVM was deployed to model fine-grained decision boundaries with contextual flexibility. Using the study of Jain & Kaur, (2017) as inspiration, this hybrid approach adapts to uncertainty and tactical variability in team performance, particularly across transitional points in the season. The fuzzy component enhances model sensitivity to ambiguous or borderline cases, while the SVM portion of the model guarantees vigorous separation in the feature space. This dual-character model was especially effective for modelling competitive volatility and behaviour near decision thresholds.

XGBoost was integrated for its strength in structured learning and its compatibility with SHAP-based interpretability. The model was configured to optimise precision while maintaining resistance to overfitting through shrinkage, regularisation, and boosting iteration control. SHAP value analysis was used to assess feature contributions across different competitive contexts, providing actionable insights into how input variables influenced predictions in regular versus postseason matchups. XGBoost played a key role in both predictive accuracy and diagnostic interpretation, and is a main model used in past literature.

The FNN architecture was constructed with two hidden layers (64 and 32 neurons), ReLU activation, and dropout regularisation, the network was trained with binary cross-entropy loss and the Adam optimiser. Min-max scaled inputs fed into the sigmoid output provided binary win predictions. Early stopping callbacks were applied to prevent overfitting. The model demonstrated strong pattern learning capacity and served as a complementary perspective to tree-based and margin-driven classifiers and was derived from the ANN model used in the study by Thabtah et al., (2019) in where training conditions are repetitively adjusted until the wanted yield is attained.

To synthesise the strengths of multiple classifiers, the ensemble modelling approach was adopted, which was inspired by the work of Cai et al., (2019). In that study, a hybrid ensemble framework trained with bagging and random subspace routine for basketball results projection. Fusion logic combining Random Forest and SVM predictions supported with fuzzy weighting schemes. This strategy utilised the stability of RF, and combined it with the precision of SVM logic, and the

adaptability of fuzzy rule modelling. Ensemble outputs offered smoother prediction boundaries and improved generalisation performance, especially in contested matchups and transitional phases between regular season and playoffs.

3.5 - Evaluation & Visualisation

Model performance was assessed using a complement of standard classification metrics, including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUC). These were computed across each competitive context to assess sensitivity to game type and feature interactions. Cross-validation was used to ensure generalisability, and permutation testing was applied for strength analysis.

To assess calibration accuracy and interpret alignment between predictions and outcomes, team-level scatter plots of fitted P_e values versus actual win percentages were generated. These logo-overlaid plots were constructed for each season independently and provided intuitive diagnostics for model fidelity, outlier behaviour, and season-over-season volatility. Feature importance tables, residual plots, and matchup-type confusion matrices were compiled to support comparative analysis across all models. These visual and statistical diagnostics guided final model selection and helped narrate where predictive precision was strongest and where model sensitivity faltered.

Chapter 4.0 - Results

This section displays the core outcomes of the study by evaluating the performance of P_e and ML models. The aim is to assess how each model type performs under varying competitive pressures and data environments. Evaluation metrics include F1 score, accuracy, precision, recall, and ROC AUC, RMSE, MAE and R^2 Scores.

Prior to implementing any P_e model, optimal exponents were independently calibrated for each regular season, with separate postseason values derived to account for playoff-specific scoring dynamics (see Table 3).

Table 3: Seasonal exponent estimates using Fitted & Weibull derived Pe models in the regular and post season contexts.

Season	Regular (Fitted)	Regular (Weibull)	Post (Fitted)	Post (Weibull)
2022	12.8403	18.2713	15.0866	19.0249
2023	14.2232	19.1752	17.8466	19.8888
2024	14.2742	18.6488	12.7721	19.8209
2025	13.2862	18.9061	11.4739	17.4898

Table 3 presents fitted and Weibull-derived Pythagorean exponents for regular and postseason NBA data across four seasons. Regular season ‘fitted’ values remain stable, ranging from 12.84 to 14.27, while Weibull estimates are consistently higher (18.27–19.18). In contrast, postseason fitted exponents show a sharp decline, peaking at 17.85 in 2023 and falling to 12.77 in 2024.

Figure 3: Estimated Pythagorean Exponents by Season (2022–2025): Fitted vs Weibull-Derived Pe Models.

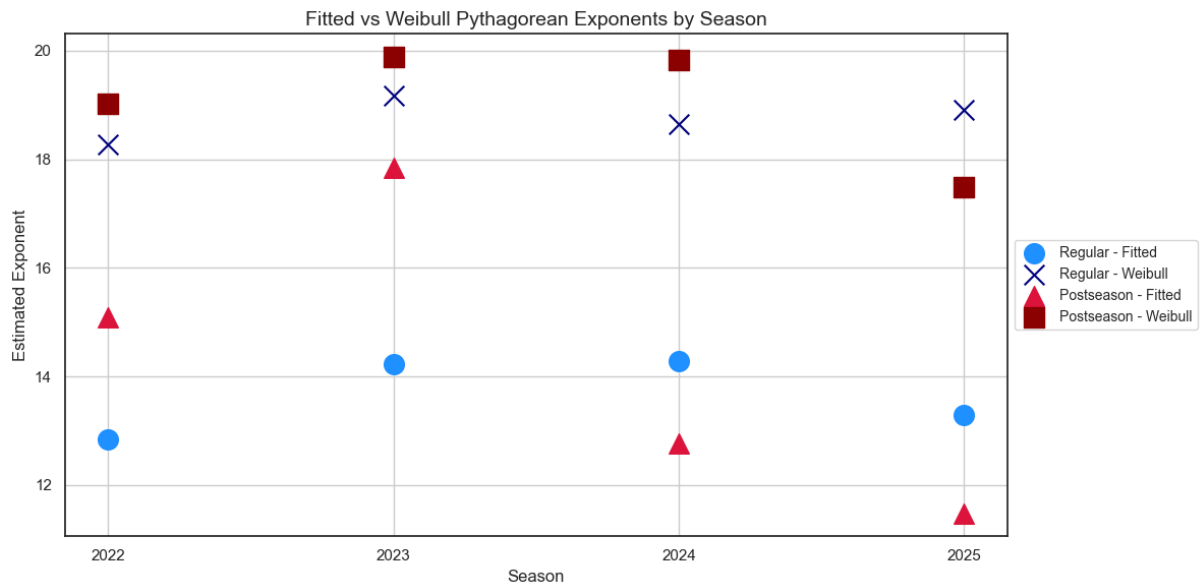


Figure 3 demonstrates the contrast between ‘fitted’ and Weibull-derived Pythagorean exponents across regular and postseason games from 2022 to 2025. The decline in the ‘fitted’ exponent for playoffs between 2023 and 2024 indicates that point differential becomes a less reliable predictor of win percentage. The Weibull-derived exponents remain consistently higher, suggesting that theoretical scoring shapes imply greater discriminative power than what is observed in postseason outcomes.

4.1 - Performing Pe

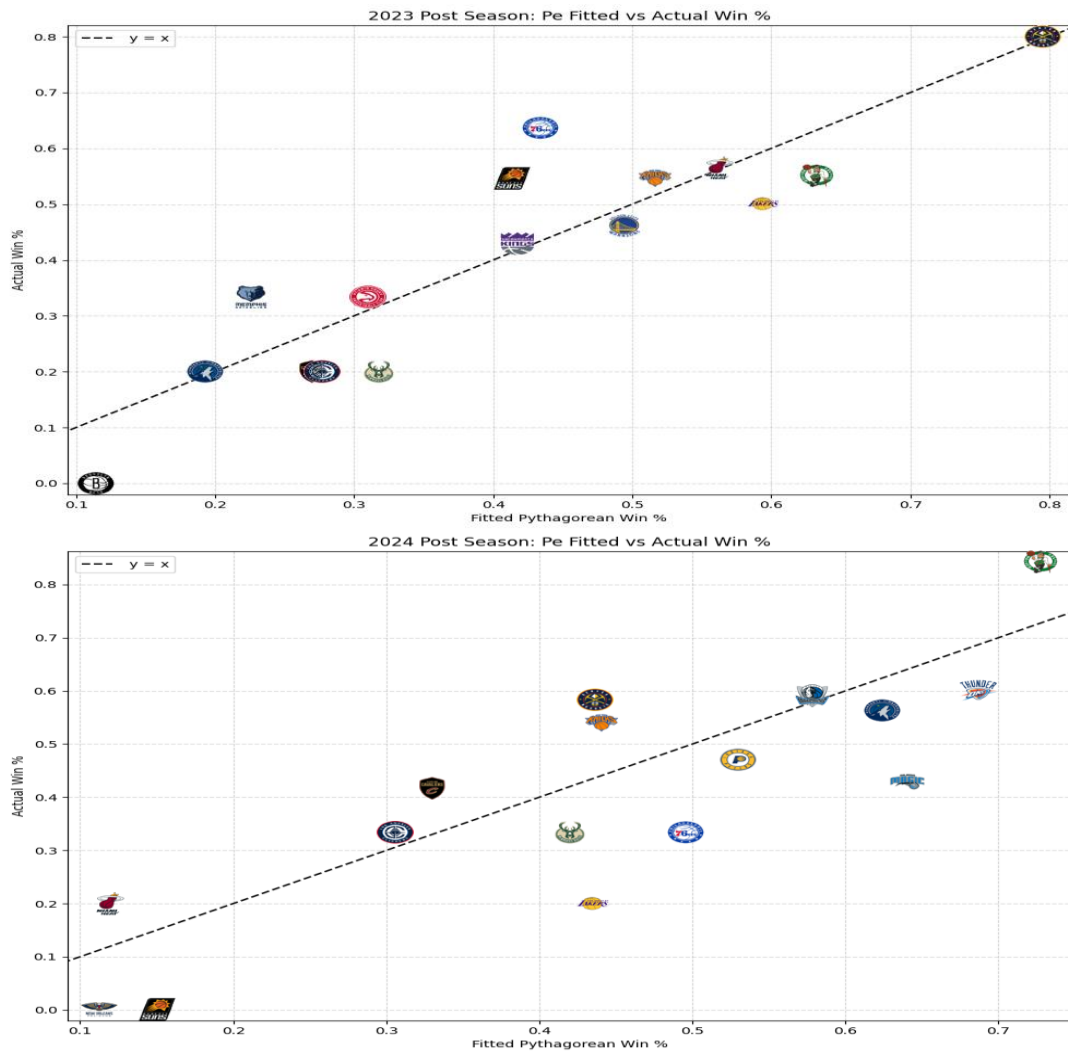
With the optimal Pythagorean exponent now calibrated across each season, the Pe model could be formally executed to generate expected win percentages for each team. These exponents were applied to team-level aggregates of points scored and allowed, following the standard Pe formula stated by Oliver, (2003), seen in Figure 4.

Figure 4: *Pythagorean Expectation Formula for Team Win Percentage Estimation.*

$$\text{Win}\% = \frac{\text{PTS}^z}{\text{PTS}^z + \text{DPTS}^z}$$

With the formula outfitted with the calculated exponents (z), the model can transform raw scoring inputs into expectation-based outputs. The postseason exponent estimates highlighted in Table 3 reveal a substantial year-on-year deviation between 2023 and 2024, with the fitted values indicating notable shifts in the statistical relationship between team scoring margins and win probabilities. Figure 5 visualises this comparison by plotting actual win percentages against Pe-derived expectations for the 2023 and 2024 postseasons respectively. Teams that lie close to the diagonal reference line exhibit strong alignment with the Pe model, while those that diverge highlight underlying strategic or statistical anomalies.

Figure 5: Postseason Pe Model Residuals: Actual Win% vs Predicted Win% in 2023 & 2024.



The comparative scatter plots in Figure 6 build directly on the residual analysis introduced in Figure 5, offering a more granular view of how point differential drives the Pe model's predictive behaviour. In Figure 6, the x-axis captures point differential, which is a core input for calculating the Pythagorean exponent. The figure highlights how varying exponent values influence prediction sensitivity, with steeper exponents amplify the impact of high-margin games, while lower exponents produce flatter estimates and broader prediction bands. This can be seen in the figure, as the 2023 postseason shows a much tighter range of points differential than in 2024, which is far more spread.

Figure 6: Postseason Win% vs Point Differential: Pe Model Comparison Across Seasons 2023 & 2024.

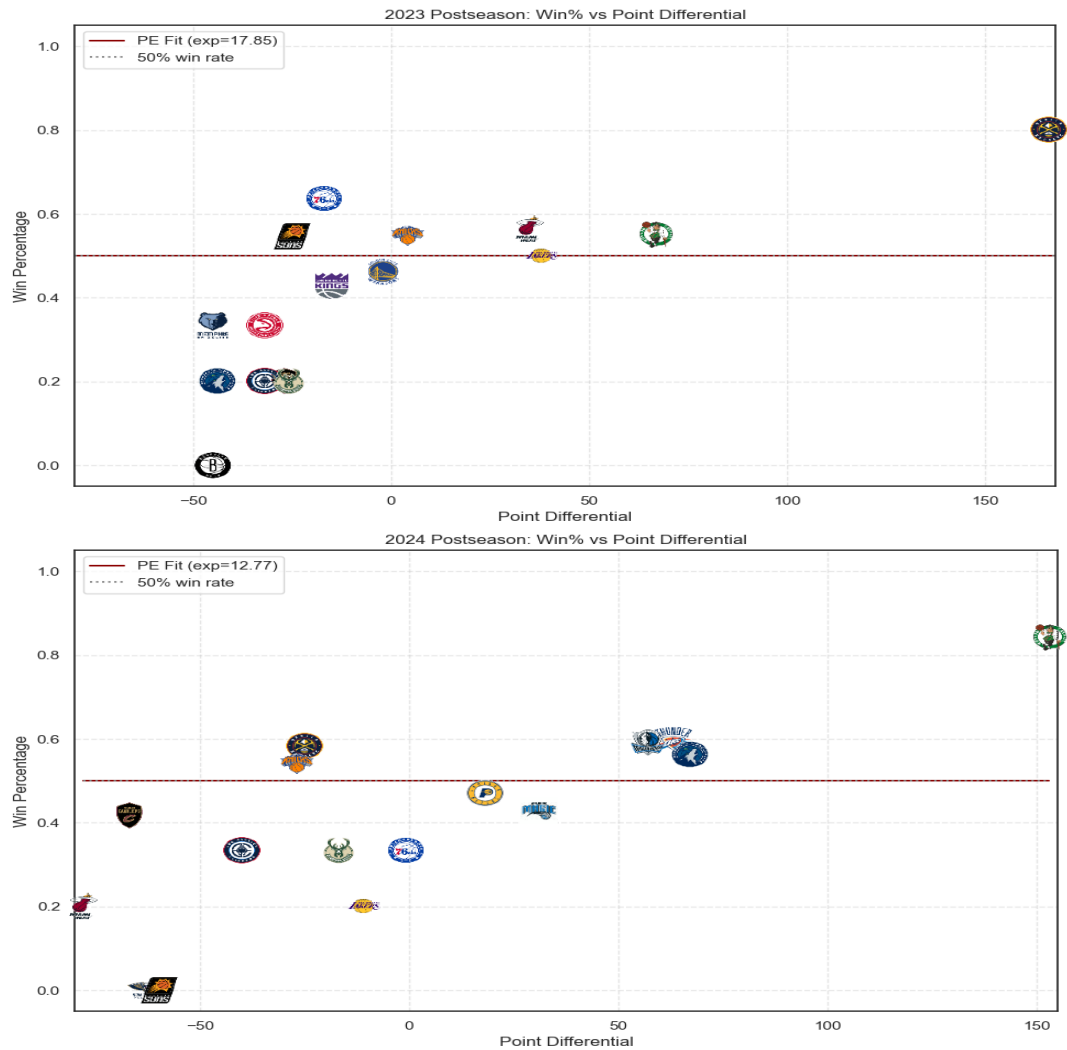


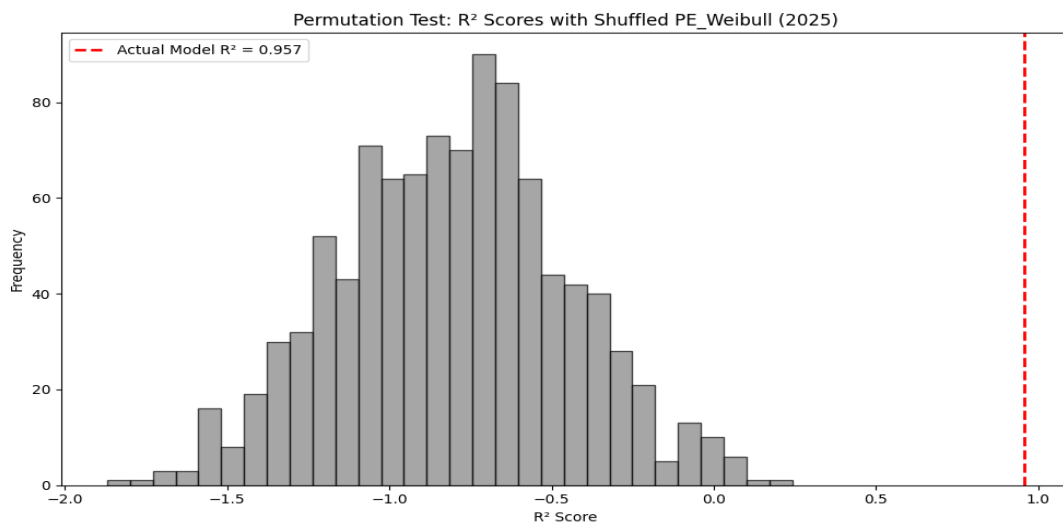
Table 4 highlights the degree of alignment between model estimates and observed performance, offering insight into the statistical fidelity of the Pe approach. Metrics such as mean absolute error (MAE), root mean squared error (RMSE) were computed to quantify model accuracy. The results exhibit that while the Pe model maintains strong predictive consistency in regular season contexts, postseason volatility introduces greater residual spread. This volatility undermines the Pe model’s assumptions of consistent scoring logic and smooth performance distributions.

Table 4: Validation of Pythagorean Expectation Model: Predicted vs Actual Win Percentages Across Seasons.

Model	RMSE	MAE	R ²
Fitted Regular Season	0.0416	0.0329	0.9326
Weibull Regular Season	0.0331	0.0266	0.9572
Fitted Post Season	0.0920	0.0801	0.8111
Weibull Post Season	0.0969	0.0838	0.7903

To assess the reliability of the Weibull-derived Pe model, a permutation test was conducted (Figure 7). The procedure involved randomly shuffling predictor values in the 2025 test set across 1,000 iterations, each time recalculating the R² score using the original trained model. This generated a null distribution representing performance levels expected under the assumption of no true predictive relationship. The actual R² of the model fell well outside this baseline range, confirming that the observed predictive accuracy is not attributable to random chance.

Figure 7: Permutation Test Distribution of R² Scores: Validating ‘PE_Weibull’ Feature Impact (2025 Regular Season).



The Pe model has performed effectively in regular season scenarios, however a consistent decline of approximately 10–15% in predictive accuracy is observed when applied to postseason matchups. A primary factor is the substantially smaller sample size of playoff games, which limits the statistical stability of exponent fitting and win percentage estimations. With fewer observations available, individual anomalies and outlier performances carry greater influence, amplifying residual noise and

weakening generalisability. To improve the model’s robustness under postseason conditions, teams were retrospectively flagged during the regular season using playoff qualification markers. This enabled the creation of matchup-type classifications, thereby simulating postseason intensity across a more expansive dataset (Table 5).

Table 5: Contextual Model Validation: Comparative Pe Performance Across Matchup Types.

Context	Model	RMSE	MAE	R ²
Non-Playoff vs Non-Playoff	Fitted	0.0571	0.0436	0.8223
Non-Playoff vs Non-Playoff	Weibull	0.0774	0.0630	0.6742
Playoff vs Non-Playoff	Fitted	0.0518	0.0421	0.9518
Playoff vs Non-Playoff	Weibull	0.0648	0.0526	0.9245
Playoff vs Playoff	Fitted	0.0484	0.0397	0.7680
Playoff vs Playoff	Weibull	0.0595	0.0491	0.6492

This segmentation allowed the Pe model to simulate postseason intensity with greater granularity. Across all contexts, the ‘Fitted’ model consistently outperforms the Weibull-derived variant, demonstrating lower RMSE and MAE values and stronger R² alignment. Notably, a decline in R² and a rise in error metrics is observed in the ‘PvP’ row for both models. These limitations highlight the potential benefits of deeper ML architectures that incorporate richer feature sets and capture non-linear relationships. ML models can surpass the ceiling imposed by single-metric predictors like point differential with the capacity to absorb more granular context. These more expressive models offer a compelling pathway toward enhanced predictive performance.

4.2 - ML Applications

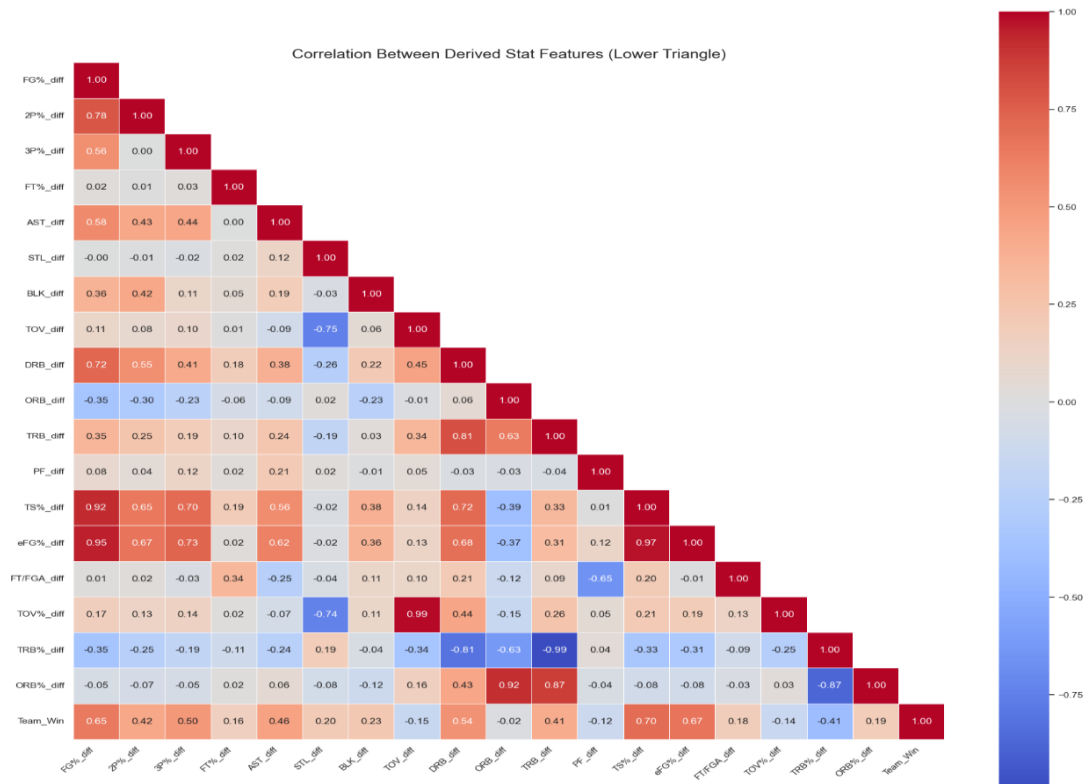
ML models are designed to learn from multidimensional input data, allowing for complex interactions between scoring metrics, opponent strength, game context, and team-specific tendencies. Examining how these models leverage feature importance, which is a measure of how influential each variable is in determining predicted outcomes and to uncover the drivers of team performance. Understanding feature importance enhances diagnostic insight into the model’s internal logic, helping validate whether predictions are grounded in meaningful basketball characteristics or noise-prone artifacts (see Table 6).

Table 6: Features initially selected through the feature importance process.

Shooting		Offense		Defence	
FG%	FT%	TRB	ORB	STL	BLK
2P%	3P%	TRB%	ORB%	PF	DRB
TS%	eFG%	AST	TOV		
FT/FGA			TOV%		

The correlation matrix in Figure 8 served as an analytic tool for refining feature relevance ahead of deeper ML modelling. To produce the “_diff” feature set, each individual team’s per-game statistic was calculated as a net differential by detracting the opponent’s matching feature value, creating context-adjusted metrics that reflect relative performance rather than absolute totals. Notably in Figure 5, ‘TS%_diff’ and ‘eFG%_diff’ exhibited exceptionally high correlations with winning outcomes. While their predictive strength is undeniable, these metrics are outcome-dependent that are tightly coupled with final game results. Their inclusion would risk bleeding target signals into the model, artificially inflating predictive performance. Despite their intuitive relevance, both features were deliberately excluded from subsequent model usage.

Figure 8: Correlation Matrix of Context-Adjusted Performance Metrics: Evaluating Feature Relationships and Relevance to Team Wins.



Further inspection of feature relationships revealed multiple pairs of variables serving near-identical analytical functions. Metrics such as 'TOV' and 'TOV%', 'ORB' and 'ORB%', and 'TRB' and 'TRB%' were shown to be statistically redundant, with the percentage-based versions offering stronger correlations to team success. In each case, the raw-count counterpart was omitted to minimise dimensional duplication and reduce noise. A similar logic applied to 'FG%' and '2P%', which displayed overlapping informational patterns, which prompted the exclusion of '2P%'. This targeted pruning of the feature set ensures that the ML models that are deployed avoid inflated results from leak-prone or duplicate metrics.

A range of ML algorithms were implemented to model win outcomes using context-adjusted performance features. Each approach followed a consistent pipeline of preprocessing, model fitting, and performance evaluation, with feature importance rankings used to assess variable influence. Evaluation was directed separately across full-season, postseason, and playoff-calibre subsets, with feature relevance tracked consistently via normalised importance tables (see Table 7).

Table 7: *Aggregated Feature Ranking Across Machine Learning Models: Contextual Stat Importance in Predicting Win Outcomes.*

Feature	Average ML Ranking
FG%_diff	3.38
DRB_diff	4.48
3P%_diff	4.71
TRB%_diff	4.71
TOV%_diff	5.52
PF_diff	6.67
ORB%_diff	6.95
FT%_diff	7.57
FT/FGA_diff	7.76
AST_diff	7.86
STL_diff	8.29
BLK_diff	10.10

Tables 7 and 8 present aggregated rankings of performance metrics based on their average importance across a variety of models. Each feature is evaluated for its predictive influence in estimating team win outcomes. Across all contexts (Table 5), 'FG%_diff' emerges as the most dominant predictor, with 'DRB_diff', '3P%_diff', and 'TRB%_diff' form a consistent secondary cluster of highly impactful indicators. Features 'STL_diff' and 'BLK_diff' fall near the bottom of the overall ranking, indicating inconsistent influence across game types.

When isolating playoff-calibre scenarios (Table 8), the same trends persist, 'FG%_diff' remains the top feature, reaffirming its value in tightly contested matchups. Meanwhile, '3P%_diff', 'TRB%_diff', and 'DRB%_diff' continue to show high relevance.

Table 8: Aggregated Feature Ranking Across Machine Learning Models: Contextual Stat Importance in Predicting Win Outcomes in 'Playoff vs Playoff' context.

Feature	Average ML Ranking
FG%_diff	3.14
3P%_diff	5.00
TRB%_diff	5.14
DRB%_diff	5.29
TOV%_diff	6.14
ORB%_diff	6.57
FT/FGA_diff	6.71
FT%_diff	7.14
PF_diff	7.14
STL_diff	7.71
BLK_diff	8.57
AST_diff	9.43

Table 9 summarises the predictive performance of each ML model on regular season data using 2025 as a test set. The FNN stands out with the strongest performance across all metrics with an accuracy of 0.9335 and an AUC of 0.9853. XGBoost follows closely, while the HFSVM performs least reliably, trading precision for high recall. These results reinforce the value of expressive architectures in modelling the statistical stability found within the regular season.

Table 9: Regular Season Model Performance Metrics (2025 Validation).

Context	Accuracy	ROC AUC	Precision	Recall	F1 Score
LR	0.9216	0.9789	0.9216	0.9216	0.9216
RF	0.9258	0.9769	0.9261	0.9254	0.9258
NB	0.8482	0.9789	0.8463	0.8508	0.8486
HFSVM	0.7196	0.7736	0.6437	0.9833	0.7781
XGBoost	0.9304	0.9818	0.9281	0.9330	0.9306
FNN	0.9338	0.9853	0.9317	0.9300	0.9335
Ensemble	0.9216	0.9798	0.9197	0.9239	0.9218

Table 10 presents performance under postseason-specific conditions. XGBoost leads across the board with an accuracy of 0.9345, reflecting its robustness in modelling games with tighter margins. FNN trails slightly but maintains strong generalisation. In contrast, NB and HFSVM struggle to maintain precision due to the elevated volatility of playoff games.

Table 10: Post Season Model Performance Metrics (2025 Validation).

Context	Accuracy	ROC AUC	Precision	Recall	F1 Score
LR	0.8929	0.9596	0.8929	0.8929	0.8929
RF	0.8869	0.9624	0.8824	0.8929	0.8876
NB	0.7976	0.8861	0.8125	0.7738	0.7927
HFSVM	0.6429	0.8099	0.5833	1.0000	0.7367
XGBoost	0.9345	0.9772	0.9195	0.9524	0.9357
FNN	0.9167	0.9840	0.9167	0.9167	0.9167
Ensemble	0.9167	0.9762	0.9167	0.9167	0.9167

Table 11 presents the model performance results when applied specifically to ‘PvP’ contexts. The Ensemble model delivers the greatest overall performance, including an accuracy of 0.9107 and an F1 score of 0.9104 indicating its effectiveness in navigating tactical intensity and parity between elite teams. FNN, RF, and XGBoost also show high reliability across metrics, each scoring above 0.89 in accuracy. HFSVM is an outlier in this context, showing perfect recall but severely compromised precision and balance, which implies over-sensitivity to win signals with reduced discriminative control.

Table 11: *Playoff Vs Playoff Model Performance Metrics (2025 Validation).*

Context	Accuracy	ROC AUC	Precision	Recall	F1 Score
LR	0.8810	0.9628	0.8810	0.8810	0.8810
RF	0.9062	0.9648	0.9075	0.9048	0.9061
NB	0.8250	0.9113	0.8227	0.8286	0.8256
HFSVM	0.6354	0.8118	0.5783	1.0000	0.7328
XGBoost	0.8988	0.9701	0.8941	0.9048	0.8994
FNN	0.9033	0.9778	0.8883	0.9226	0.9051
Ensemble	0.9107	0.9732	0.9137	0.9071	0.9104

Chapter 5.0 – Discussion

The central objective of this study was to evaluate whether ML classifiers trained on basketball metrics could surpass the Pe model in predicting win outcomes across a variety of competitive environments. By constructing an ensemble framework and embedding matchup classification logic, the project sought to create a reproducible, feature-driven system for win forecasting that remained effective across all game types.

5.1 – Interpretations of Key Findings & Comparison to Literature

The study’s results demonstrate that predictive accuracy in basketball is highly sensitive to feature selection and contextual game classification. Across all ML models and competitive domains, ‘FG%_diff’ and ‘3P%_diff’ consistently surfaced as the most influential predictors, echoing the real-

world intuition that shooting efficiency is the cornerstone of game success. These findings are supported quantitatively through normalised importance rankings (Tables 7 and 8), and qualitatively through prior studies such as Su and Chen, (2022) who emphasised the importance of shooting efficiency metrics. These results provide validation that shooting-centric features drive team success. Furthermore, Thabtah et al., (2019) identified 'DRB' and 'TRB%' as leading indicators when filtered through rule induction algorithms. This is significant for coaches for roster construction, training methods and in the terms of professional sports what individuals are more financially significant to organisations than others.

The contrast between regular season and playoff model performance further substantiates the conceptual divide noted in Huyghe et al., (2021) and Mandic et al., (2019), who observed statistical compression and strategic volatility in playoff contexts. In this study, ensemble models, particularly those combining fuzzy logic and tree-based classifiers emerged as the most adaptable tools in 'PvP' matchups. This meant that the model showed a greater ability to navigate compressed feature distributions and tactical disruptions than linear baselines. This behaviour aligns with the hybrid frameworks presented by Cai et al., (2019) who found that when you initialise multiple ML algorithms, the ability to find greater distinction in the marginal factors that contribute to winning outcomes increases exponentially. Furthermore, this is an improvement over Pe models, which showed persistent degradation in predictive power across postseason segments (see Table 5).

The application of Weibull-derived exponent estimation proved more stable but revealed less discriminative precision when applied to volatile playoff contexts, reinforcing the limits of distribution-based modelling when confronted with nonstationary tactical environments, echoing the challenges identified by Almeida et al. (2024). To address the scarcity of postseason data, simulated playoff-calibre matchups enabled a broader feature representation under elevated tactical pressure. This extension builds on Michailidis et al. (2025)'s framework on real-time predictive Pe calibration, allowing models to capture playoff-specific compressions in feature dynamics with improved sensitivity. Notably, the integration of adaptive features aligns with Lu et al., (2021), who demonstrated that using weighted game-lag information alongside ensemble ML techniques significantly improves prediction accuracy. These integrated methods validated this study's findings that ensemble ML frameworks attuned to contextual volatility are uniquely

equipped to handle high pressure environments, where conventional distributional approaches often fail to capture decisive performance shifts.

These findings validate the use of expressive models trained on well-engineered differential inputs over the use of formula-driven baselines in volatile game contexts, while also grounding the results in a broader body of academic research that supports adaptive predictive design. Building on the foundation of Yuan et al., (2015), this study reinforces the methodological merits of adaptive ML modelling. Yuan's ensemble strategy for NCAA tournament prediction reveals the efficacy of diverse model architectures when trained on rigorously decontaminated features. Similarly, Turner and Franks, (2021) synthesises two decades of evolving practice, highlighting how expressive models can outperform rigid baselines by capturing latent variance across player, team, and strategic dimensions. Both studies support a wider shift in basketball analytics toward using richer data, smarter evaluation techniques, and models that reflect the real complexity of the sport in the professional context.

5.2 – Theoretical & Conceptual Linkages

By employing context-adjusted differentials and matchup-driven segmentation, this study contributes to a dynamic modelling framework that handles team performance as relational to the opposition they are facing. This aligns with the methods outlined in Ma et al., (2023), which demonstrated that traditional cumulative metrics and static team-level features can underperform in predictive tasks. While baseline models offered useful benchmarks, the study's more advanced implementation of XGBoost enriched by robust feature engineering markedly improved accuracy. In contrast to static averages, this study's engineered differentials captured comparative team strength, which offered a more detailed representation of situational dynamics. This builds upon Ma's emphasis on feature validity and robustness, advancing the case that relational metrics better reflect game volatility and enhance postseason predictions.

Predictive volatility revealed through diminished model reliability in postseason and 'PvP' matchups, can be understood as a manifestation of tactical fluidity and psychological variability in elite-level competition. Unlike regular season games, playoff-calibre matchups often feature strategic adaptation, minimal lineup experimentation, and defensive prioritisation causes disruption to traditional scoring flows. The findings resonate with Ke et al., (2024) who emphasised the need for

reduced dimensionality and targeted KPI extraction to tame noise and preserve signal relevance in roster optimisation settings. The strength of ensemble and FNN models rests in their capacity to absorb latent patterns regardless of competitive environment. Their layered structure enables the capture of abstract representations such as playstyle shifts, fatigue effects, or tactical rotations that simpler linear models fail to anticipate.

This is particularly relevant given the performance of the FNN model in this study, where prediction accuracy was constrained by overfitting, even after dropout and early stopping techniques were applied. Nevertheless, the link between depth-oriented architectures and the need for temporal sensitivity remains intact. The model's ability to encode subtle temporal fluctuations, though imperfect in execution emphasised the nonlinear interactions that are inherent in NBA games. When aligned with the present study's it becomes evident that depth-based architectures might not outperforming all models, but still offers theoretical promise for capturing in-game volatility and postseason complexity.

5.3 – Research Limitations & Reasonings

Several factors may explain the observed discrepancies in model performance across competitive segments. The drop in predictive reliability within postseason matchups may stem from changes in team behaviour, including heightened defensive intensity, reduced rotational depth, and strategic pacing. These are all variables that are not explicitly captured by box score features but capture the nuanced unpredictability that can compromise model efficiency and amplifies feature instability. Furthermore, while the tagging logic used to simulate 'PvP' intensity within regular season games expanded the dataset meaningfully, it remains a proxy approximation. Teams often alter their tactical focus when facing fellow playoff qualifiers, and this cannot be fully replicated through record-based classification alone. This method lacks further depth into the background of heightened stakes in playoff series, where the repetitive structure of playing one team back-to-back a potential seven times can have a profound effect on the ability to accurately model outcomes.

While the elevated predictive accuracy reported in the study is an encouraging sign of methodological progress, it may also signal a potential limitation tied to model generalisation and overfitting risk. Additionally, the differential features assume consistent interpretive value across teams and matchups, potentially masking stylistic variation or anomalous play that deviates from

average behaviours. Expressive architectures like neural networks and ensemble classifiers are powerful but also inherently less transparent, making the arrival to the outcome difficult to ascertain. For example, Sikka and Devarajan, (2022) found that aggregate decision trees and gradient boosting achieved notable accuracy that is comparative to this studies results. However, their reliance on aggregate season-level inputs and point differentials risks blurring the unique situational dynamics that influence in-game volatility, especially within tightly contested or postseason contexts. Therefore, while heightened accuracy is a methodological strength, it must be carefully monitored against the risk of inflated performance metrics that don't fully account for the volatility and strategic nuance of real-world playoff environments.

Another limitation of this study lies in its exclusive focus on team-level metrics, which inherently overlooks the dynamics driven by individual player contributions. Critical in-game factors that can be attributed to specific players such as clutch shooting, defensive versatility, or elite playmaking are diluted when averaged across an entire team. This restricts the models' ability to account for match-defining moments often dictated by single-player impact, particularly in the high stake's games in the postseason. Moreover, tactical adjustments that hinge on star-player usage or bench depth are lost in aggregated data. Prior work by Papageorgiou et al., (2024) illustrates that integrating player-level KPIs through methods like clustering, game-lag evaluation, or positional analysis significantly enhances model reliability and outcome specificity. The present study provides a robust but surface-level forecast, lacking sensitivity to the individual variability that often determines playoff success. These limitations underscore the need for richer feature sets that incorporate situational or temporal indicators that are applicable to the individual as well as to a whole team or league which includes the likes of player fatigue, injury impact, or coaching strategy variance.

Additionally, there is an ethical dimension to be considered. Predictive modelling at this level of detail carries implications for roster decisions, player valuation, and strategic planning, potentially influencing everything from trade negotiations to contract structuring. If models prioritise high-variance or high-volume metrics such as points scored, usage rate, or shot attempts, this can inadvertently bias evaluations toward specific player types, such as isolation scorers or primary ball-handlers, while undervaluing traits that lack easier quantification. Intangible elements like leadership presence, off-ball movement, adaptability to changing game plans, or chemistry-driven

synergy between teammates lack statistical categorisation but remain vital to competitive success. Moreover, reliance on opaque architectures like ensemble models or deep neural networks can obscure these biases, leaving decision-makers unaware of what trade-offs are being baked into predictive outcomes. As ML techniques deepens its influence in professional sport, safeguarding against reductive or overly mechanistic interpretations becomes essential to uphold methodological integrity but also to honour the complexity of the human elements at play.

5.4 – Implications for Practice

The findings of this analysis offer tangible intuitions for basketball analysts, coaching staff, and organisational decision-makers seeking strategies for performance evaluation and game preparation. The consistent prominence of differential metrics, particularly those related to shooting efficiency and rebounding suggests that analysts should prioritise efficiency-based indicators over traditional aggregate statistics when assessing competitive advantage. While these findings largely reaffirm existing literature on the importance of rebounding and shooting proficiency, particularly the work of (Thabtah et al., 2019; Turner and Franks, 2021; Ke et al., 2024), which illustrates the emphasis needed on contextual suppression of opponent strengths as a pathway to success. Specifically, this study identifies defensive rebounding differential ('DRB') as a key predictive feature, with teams outperforming their opponents in 'DRB' exhibited a 75% likelihood of achieving a successful outcome. This underscores the strategic value of limiting second-chance opportunities, an insight that may warrant greater emphasis than previously reported in similar analyses.

The demonstrated success of ensemble methods and neural network architectures in navigating playoff volatility underscores the importance of diversified modelling frameworks in high-stakes professional sports forecasting. Postseason exhibits heightened tactical variability, lineup disruptions, and situational unpredictability factors that can destabilise certain models or approaches. By engaging a composite modelling strategy that influences the strengths of multiple algorithms, analysts can effectively mitigate forecast uncertainty while capturing nuanced tactical patterns that emerge in complex game environments. This multi-model methodology improves predictive robustness, as well as enabling dynamic scenario testing, which is critical for strategic planning across different playoff matchups. Furthermore, this study's emphasis on methodological reproducibility and modular pipeline design ensures seamless integration into existing analytics

ecosystems. This makes the framework both scalable for long-term application and adaptable to evolving data sources, supporting broader institutional adoption by professional teams and analytics departments.

5.5 – Link to Research Questions and Future Directions

This study set out to investigate whether ML models could outperform traditional statistical baselines in predicting NBA game outcomes. The results not only validate this premise across multiple model architectures and competitive segments, but also reveal critical insights into feature relevance, volatility sensitivity, and model adaptability. By anchoring predictive logic in differential statistics and expanding postseason-like coverage, the study offers both methodological clarity and practical utility.

Looking ahead, one promising direction involves the integration of player-level variability, capturing how individual contributions influence broader team dynamics, with clustering analysis to better understand how players are grouped and what composition of player types drive success. The work of Dehesa et al., (2019) laid foundational work that identified KPI's that revealed how player interactions inform their performance, and collectively the team performance. This can be coupled with the study of Ke et al., (2024) who acknowledged clusters of players, by their underlying decision-making patterns that drive outcomes. This approach would move beyond the standard statistical metrics and captures how players adapt and influence team flow, particularly in unpredictable competitive environments. By combining these approaches with the modelling pipeline of this current study, analysts could uncover synergistic player groupings and those whose combined traits consistently elevate team performance. This would enhance descriptive profiling, as well as enabling prescriptive analytics, which would guide decisions on roster construction, substitution timing, and tactical adjustments with greater precision.

Another direction is the adoption of graph-based modelling techniques, which can represent inter-team relationships and incorporate game flow, spatial positioning, or tempo adjustments, allowing studies to capture the fluid nature of basketball strategy. Zuccolotto et al., (2019) introduced a spatial statistical framework that partitioned the court into performance-optimised rectangles. This enabled the creation of spatial performance indicators (SPIs) that reflect shooting efficiency and play dynamics across each zone. Complementing this perspective, Wang et al., (2022b) explored the

use of ensemble ML models to predict game outcomes using explainable AI methods like SHAP or LIME. These can bridge the gap between predictive power and tactical decision-making. Drawing these studies offer clear improvements to this study's design. The spatial partitioning suggests replacing fixed court zones with performance-based segmentation, enabling more accurate player comparisons and tactical insights. Explainable AI tools encourage greater model transparency, helping classify which features most influence outcomes. Adopting these approaches would sharpen future studies analytical precision and enhance its value for coaching and performance decisions.

Ultimately, this research contributes a step forward in basketball analytics, shifting the paradigm from static estimators to adaptive, context-sensitive modelling frameworks that speak not just to what happened, but why it happened, and where strategic advantage lies moving forward.

Chapter 6.0 – Conclusion

This study set out to examine whether a ML framework grounded in context-adjusted performance metrics could outperform traditional statistical models in predicting win outcomes in professional basketball. The study was driven by the belief that adaptive systems capable of accounting for matchup dynamics and tactical variation would offer stronger predictive reliability than formulaic baselines, particularly in high-pressure playoff environments.

The core aim was achieved through a multi-model pipeline that engineered differential features, segmented matchup types, and implemented diverse classifiers. These models were tested on both regular season and playoff-calibre data. Results across all contexts consistently showed that ML models offer enhanced predictive accuracy and contextual sensitivity, with ensemble approaches and feedforward neural networks performing best when volatility was highest.

Each objective was addressed in full. First, a reproducible feature engineering strategy was developed using team-versus-opponent differentials, which increased interpretability and ensured model input remained context-aware. Second, a playoff segmentation mechanism was introduced by retrospectively tagging regular season matchups based on qualification status, expanding the number of postseason-style games and allowing for deeper cross-context comparison. Third, feature importance analysis identified a clear hierarchy of influential metrics, most particularly 'FG%_diff', 'DRB_diff', and 'TRB%_diff' whose dominance remained stable across competitive

domains. Finally, the study benchmarked all ML outputs against multiple versions of the Pe model, including curve-fitted and Weibull-derived variants, confirming a consistent drop in Pe reliability during playoff matchups.

The guiding research question, whether ML could generalise better than traditional approaches was answered affirmatively. However, some limitations warrant reflection. The impressive predictive gains may be by overfitting within expressive model architectures, in addition the team-level nature of the analysis excludes player-specific variability, which can significantly shape outcome forecasts, especially in matchups where individual star performances outweigh aggregate team metrics.

Looking forward, several promising directions exist. Future models could incorporate player-level KPIs, game-lag performance engineering, or lineup clustering to enhance granularity. Graph-based architectures might capture inter-team interactions and spatial dynamics more fully. Furthermore, the advancements in player tracking technology can allow for tracking intricate team tendencies in spacing, rotations and shooting windows, as well as understanding ball trajectories with player movements to predict shot likelihood, turnover risk or expected value per trip. Ultimately, the study provides a forward-thinking approach to basketball prediction, one that not only meets its objectives, but lays the groundwork for a deeper understanding of performative outcomes in basketball analytics.

Glossary & Table of Abbreviations

Full Glossary of terms is available at <https://www.basketball-reference.com/about/glossary.html>

Gtm -- Season Game Number for Team

Blank -- Empty='Home', @='Away'

Opp -- Opposition Team code

Rslt -- Win(W)/Lose(L)

Tm -- Total points scored by Team

Opp -- Opponent Team points scored

OT -- Overtime needed in game

FG -- Field Goals

FGA -- Field Goal Attempts

FG% -- Field Goal Percentage

3P -- 3-Point Field Goals

3PA -- 3-Point Field Goal Attempts

3P% -- 3-Point Field Goal Percentage

2P -- 2-Point Field Goals

2PA -- 2-Point Field Goal Attempts

2P% -- 2-Point Field Goal Percentage

eFG% -- Effective Field Goal Percentage

This statistic adjusts for the fact that a 3-point field goal is worth one more point than a 2-point field goal.

FT -- Free Throws

FTA -- Free Throw Attempts

FT% -- Free Throw Percentage

ORB -- Offensive Rebounds

DRB -- Defensive Rebounds

TRB -- Total Rebounds

AST -- Assists

STL -- Steals

BLK -- Blocks

TOV -- Turnovers

PF -- Personal Fouls

FG_opp -- Opponent Field Goals

FGA_opp -- Opponent Field Goal Attempts

FG%_opp -- Opponent Field Goal Percentage

3P_opp -- Opponent 3-Point Field Goals

3PA_opp -- Opponent 3-Point Field Goal Attempts

3P%_opp -- Opponent 3-Point Field Goal Percentage

2P_opp -- Opponent 2-Point Field Goals

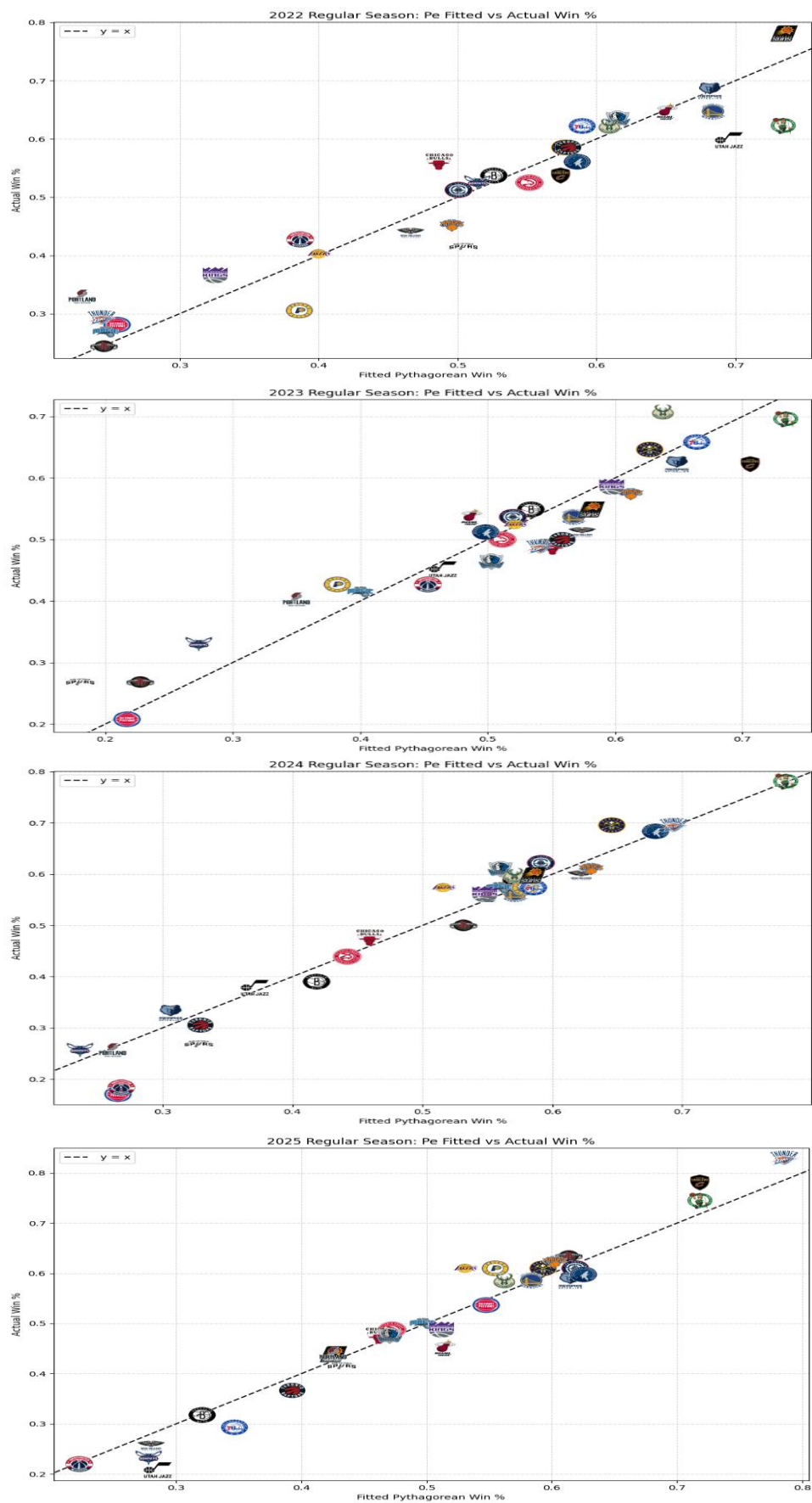
2PA_opp -- Opponent 2-Point Field Goal Attempts

2P%_opp -- Opponent 2-Point Field Goal Percentage

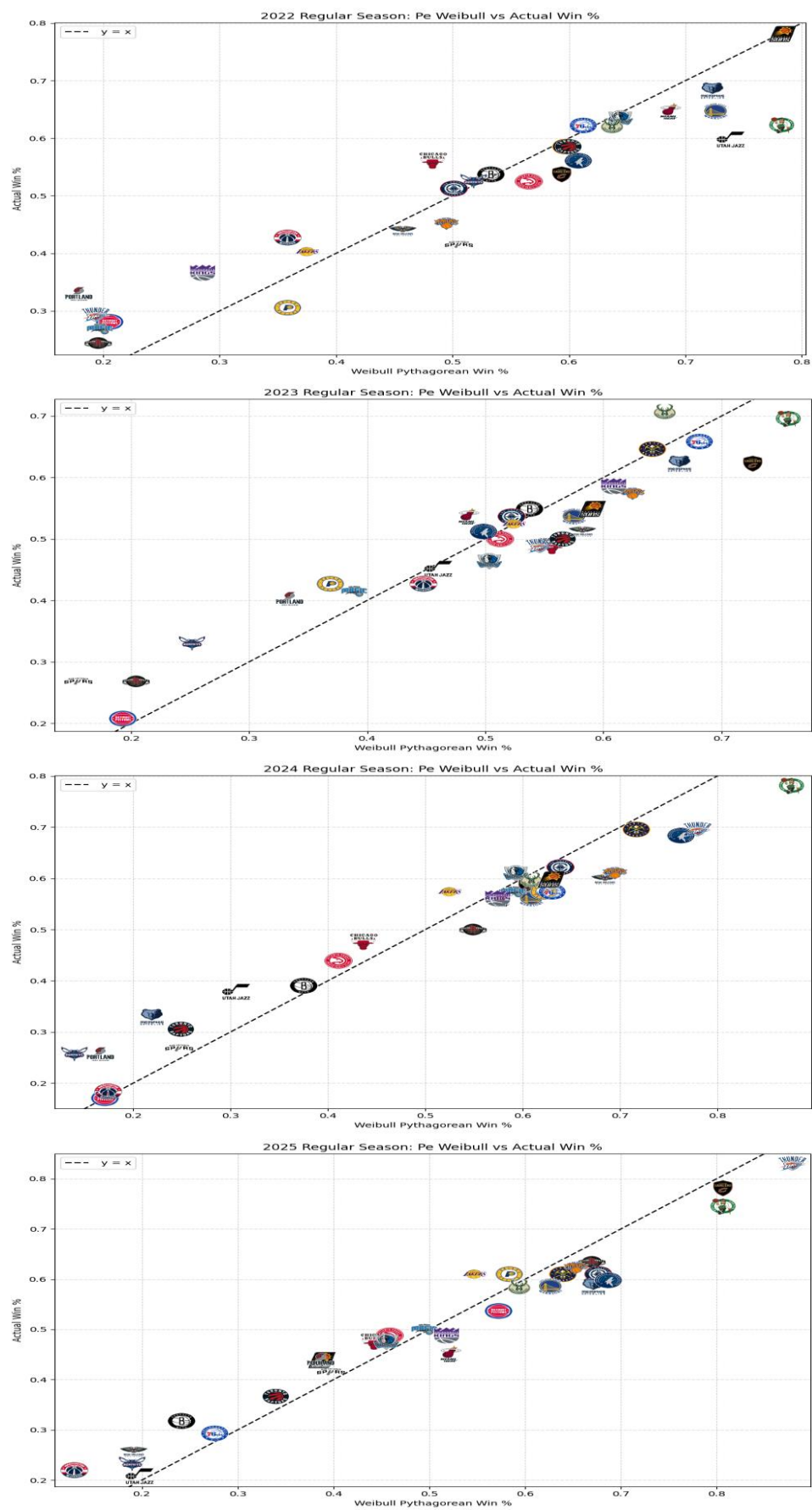
eFG%_opp -- Opponent Effective Field Goal Percentage
FT_opp -- Opponent Free Throws
FTA_opp -- Opponent Free Throw Attempts
FT%_opp -- Opponent Free Throw Percentage
ORB_opp -- Opponent Offensive Rebounds
DRB_opp -- Opponent Defensive Rebounds
TRB_opp -- Opponent Total Rebounds
AST_opp -- Opponent Assists
STL_opp -- Opponent Steals
BLK_opp -- Opponent Blocks
TOV_opp -- Opponent Turnovers
PF_opp -- Opponent Personal Fouls
ORTg -- An estimate of points produced (players) or scored (teams) per 100 possessions
DRTg -- An estimate of points allowed per 100 possessions
Pace -- **Pace Factor**: An estimate of possessions per 48 minutes
FTTr -- **Free Throw Attempt Rate** Number of FT Attempts Per FG Attempt
3PAr -- **3-Point Attempt Rate** Percentage of FG Attempts from 3-Point Range
TS% -- **True Shooting Percentage** A measure of shooting efficiency that takes into account 2-point field goals, 3-point field goals, and free throws.
TRB% -- **Total Rebound Percentage** An estimate of the percentage of available rebounds a player grabbed while they were on the floor.
AST% -- **Assist Percentage** An estimate of the percentage of teammate field goals a player assisted while they were on the floor.
STL% -- **Steal Percentage** An estimate of the percentage of opponent possessions that end with a steal by the player while they were on the floor.
BLK% -- **Block Percentage** An estimate of the percentage of opponent two-point field goal attempts blocked by the player while they were on the floor.
TOV% -- **Turnover Percentage** An estimate of turnovers committed per 100 plays.
ORB% -- **Offensive Rebound Percentage** An estimate of the percentage of available offensive rebounds a player grabbed while they were on the floor.
FT/FGA -- Free Throws Per Field Goal Attempt
TOV%_opp -- Opponent Turnover Percentage
ORB%_opp -- Opponent Offensive Rebound Percentage
FT/FGA_opp -- Opponent Free Throws Per Field Goal Attempt

Pe -- Pythagorean Expectation

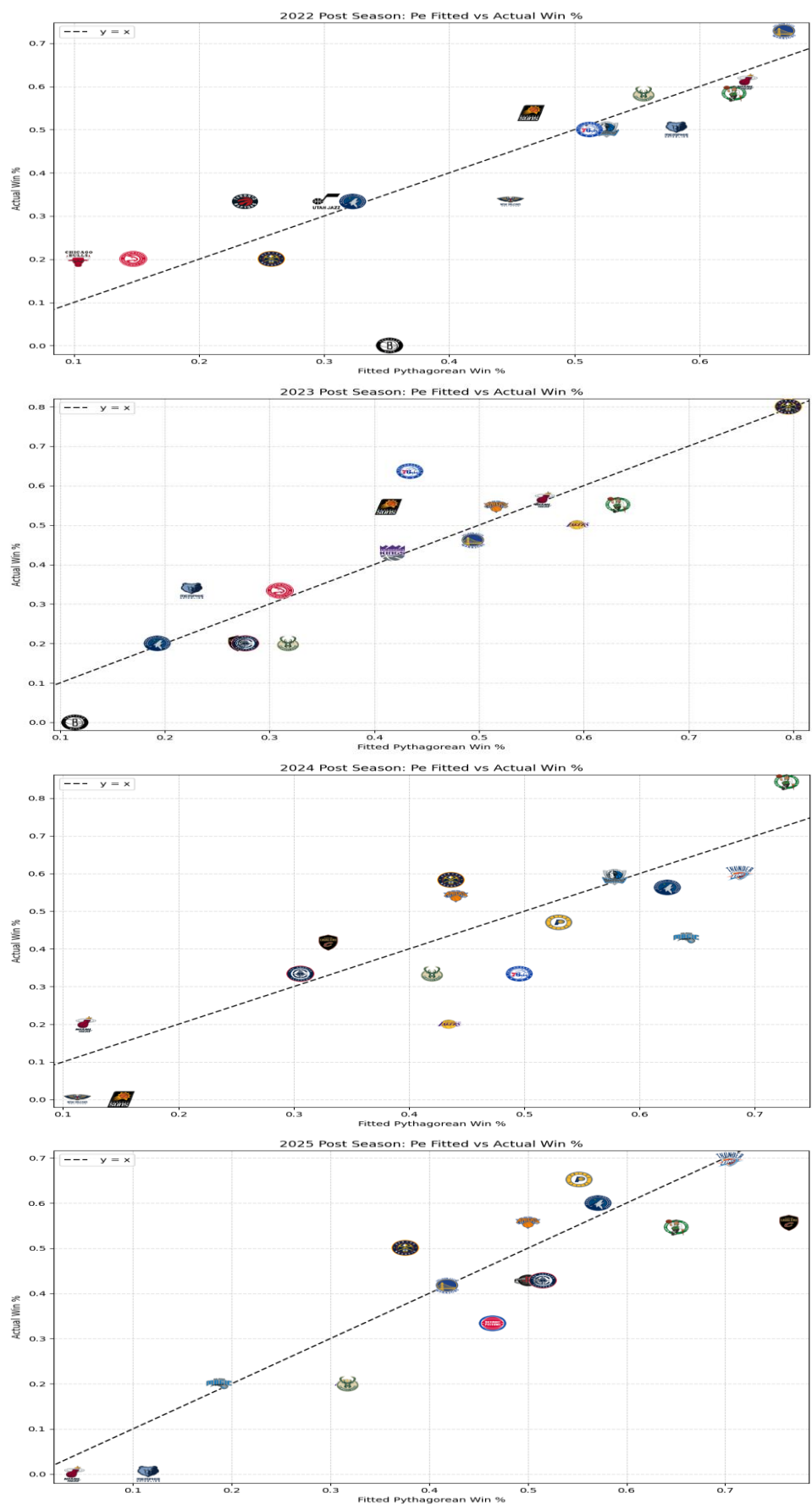
Appendix A – Pe Fitted vs Actual Win% in 2022-2025 (Regular Season)



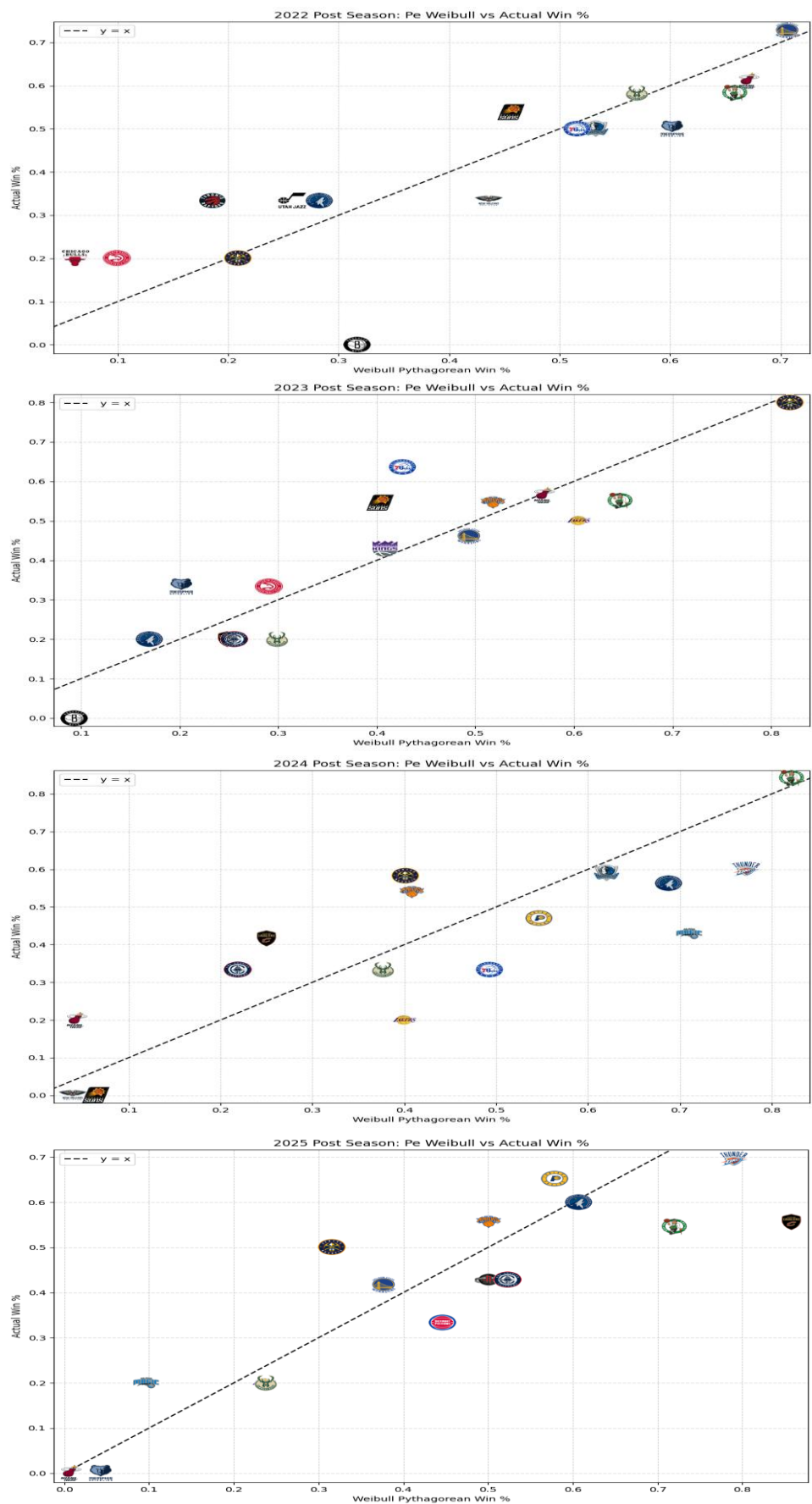
Appendix B – Pe Weibull vs Actual Win% in 2022-2025 (Regular Season)



Appendix C – Pe Fitted vs Actual Win% in 2022-2025 (Postseason)



Appendix D – Pe Weibull vs Actual Win% in 2022-2025 (Postseason)



Appendix E – Research Project Ethics Form Approval

When undertaking a research or innovation project, Cardiff Met staff and students are obliged to complete this form in order that the ethics implications of that project may be considered.

The document ***Ethics application guidance notes*** will help you complete this form and is available from the Ethics Governance Section of the Cardiff Met website. The School or Unit in which you are based may also have produced some guidance documents which you can access via your supervisor or School Ethics Coordinator.

PLEASE NOTE:

Participant recruitment or data collection MUST NOT commence until ethics approval has been obtained. PART ONE

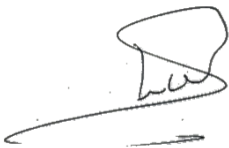
1A: GENERAL INFORMATION	
Name of applicant:	Luke Strong - CST-PG-165
Supervisor (if student project):	Ian Smith
School / Unit:	Cardiff Metropolitan University / School of Technology
Student number (if applicable):	St20121704
Programme enrolled on (if applicable):	MSc Data Science
Project Title: If using a working title, it should convey what the project is about	Beyond the Baseline: How the Use of Machine Learning Models Enhances Win Prediction in NBA Basketball
Expected start date of data collection:	22/06/2025
Approximate duration of data collection:	4 weeks
Funding Body (if applicable):	N/A
Other researcher(s) working on the project: If your collaborators are external to Cardiff Met, include details of the organisation they represent	None
Will the study involve NHS patients or staff? If yes, attach a copy of your NHS application to this form	No
Will the study involve human samples and/or human cell lines?	No

1B: Does your project fall entirely within one of the following categories:	
Desk based, involving only documents and not involving the collection of data from participants	Yes/ No
Laboratory based, not involving human participants, human samples, animals or animal derived material	Yes No /
Practice based not involving human participants (eg curatorial, practice audit)	Yes No /
Answering YES to any of these questions indicates that the project does not include any participants and you will not therefore be collecting participant data. If this is the case, please provide a short (150 words) non-technical summary of the project, complete the	
Declaration at the bottom of the form and forward this form to your School Ethics Committee (or equivalent). No further information regarding your project is required and you do not need to complete any more sections of this form. The only exception to this is in relation to projects which include security sensitive research – see below for further details. If you have answered NO to all of these questions, please proceed to 1C.	
Provide a non-technical summary of the project below. NB: Any project which includes security sensitive research will be referred to the Prevent Co-ordinator so it can be risk assessed in line with the Prevent Policy. Ethics approval will only be granted once the Prevent Co-ordinator is content with the outcomes of that risk assessment.	
This project presents a novel hybrid framework for predicting outcomes of NBA basketball games by integrating traditional statistical models with modern machine learning techniques. It combines the Pythagorean expectation—a method originally developed in baseball to estimate a team’s win percentage based on points scored and allowed—with advanced ML algorithms that capture complex, non-linear patterns in vast game data. By merging these two approaches, the model aims to improve both accuracy and interpretability, supporting better decision-making in player evaluation, roster optimisation, and overall game strategy. The framework uses historical data from recent seasons for model training and benchmarking, while real-time data is employed to test and refine predictions during the current season. In reviewing existing literature, the project identifies the strengths and weaknesses of conventional models and ML applications, thereby justifying the need for a combined, data-driven approach that delivers more robust forecasting of basketball game outcomes.	

1C: Does your project fall entirely within one of the following categories:	
Compulsory projects in professional practice (eg Initial Teacher Education)	Yes No /
A project for which NHS approval has been obtained NB If this is the case, please ensure that you submit copies of the following with this form: • any questionnaires to be used • participant consent / asset form and withdrawal form • participant information sheets • the study protocol document • the approved IRAS document	Yes No /
A project which is not compulsory in professional practice and has gained external ethics approval from a body other than the NHS. NB If this is the case, please ensure that you submit a copy of the approved ethics application with this form.	Yes/No
If you have answered YES to any of these questions, please provide a short (150 words) non-technical summary of the project and complete the rest of Part One of this form . You do not need to complete Part Two. Forward your completed form, along with any additional documents required (as indicated above) to your School Ethics Committee (or equivalent). If you have answered NO to all of these questions, please complete the rest of this form including Part Two.	
Provide a non-technical summary of the project below:	

1D: DATA COLLECTION AND STORAGE
What types of data will you collect or create?
Secondary Quantitative Statistical Data from basketballreference.com, run by SportsRadar the official stats partner of the National Basketball Association (NBA).
How will you manage access to and security of the data?

Data will be stored on OneDrive and stored behind the password protection of my laptop device. To avoid error issues and corruption data will also be stored on my removeable hard drive.	
Will the data collected be subject to the data retention protocols of any of the following bodies? • Human Tissue Authority (HTA) • Health and Care Research Wales (HCRW) • Applications involving the NHS which will be submitted via IRAS	
Yes ☐ For any project which is subject to the data retention protocols of an external body listed, you must develop a data storage plan to be submitted alongside this document for consideration by your School or Unit Ethics Panel.	
No ☒ Please confirm that the data collected will be stored in a manner which complies with Cardiff Met requirements via one of the following statements.	
STATEMENT 1: FOR STUDENTS ON TAUGHT COURSES I confirm that any non-anonymised data related to research participants will only be stored on OneDrive, or by agreement with supervising staff, on Figshare, and that all data held elsewhere will be deleted, unless it is anonymised.	☒
STATEMENT 2: FOR STAFF APPLYING ON BEHALF OF STUDENTS ON TAUGHT COURSES I confirm that all students covered by this application are aware of their obligation to ensure that non-anonymised data related to research participants must only be stored on their Cardiff Met student OneDrive account and that all data held elsewhere must be deleted, unless it is anonymised.	☐
STATEMENT 3: FOR RESEARCH STUDENTS AND STAFF I confirm that any non-anonymised data related to research participants will be stored in a secure manner (using a platform such as OneDrive or FigShare) and that all data held elsewhere will be deleted unless it is anonymised.	☐

DECLARATION: I confirm that this project conforms with the Cardiff Met Research Integrity & Governance Framework I confirm that I will abide by the Cardiff Met requirements regarding confidentiality and anonymity when conducting this project. STUDENTS: I confirm that I will not disclose any information about this project without the prior approval of my supervisor.	
Signature of the applicant:  Luke Strong	Date: 03/06/2025
FOR STUDENT PROJECTS ONLY	
Name of supervisor: Ian Smith	Date: 04/06/2025
Signature of supervisor: 	

Research Ethics Committee use only
Decision reached: Click here to enter text.
Project reference number: Click here to enter text.

Name: Click here to enter text.	Date: Click here to enter a date.
Details of any conditions upon which approval is dependant: Click here to enter text.	

PART TWO

If you haven't already done so elsewhere on this form, in the box below, provide a short (150 words), nontechnical summary of the project.	
This project presents a novel hybrid framework for predicting outcomes of NBA basketball games by integrating traditional statistical models with modern machine learning techniques. It combines the Pythagorean expectation— a method originally developed in baseball to estimate a team's win percentage based on points scored and allowed—with advanced ML algorithms that capture complex, non-linear patterns in vast game data. By merging these two approaches, the model aims to improve both accuracy and interpretability, supporting better decisionmaking in player evaluation, roster optimisation, and overall game strategy. The framework uses historical data from recent seasons for model training and benchmarking, while real-time data is employed to test and refine predictions during the current season. In reviewing existing literature, the project identifies the strengths and weaknesses of conventional models and ML applications, thereby justifying the need for a combined, data-driven approach that delivers more robust forecasting of basketball game outcomes.	
A RESEARCH DESIGN	
A1 Will you be using an approved protocol in your project?	Yes / No
A2 If yes, please state the name and code of the approved protocol to be used ¹	
N/A	
A3 Describe the research design to be used in your project In this section, include details (as appropriate) of: • Research method(s); • Sample and sampling; • Participants including recruitment methods, activities to be undertaken, time commitment, details of any proposed payments; • Analytical techniques If your project does involve the use of an approved protocol, much less details will be required but you should indicate which areas of the project are covered by the protocol.	

¹ An Approved Protocol is one which has been approved by Cardiff Met to be used under supervision of designated members of staff. For details of protocols in use in your School or Unit, contact your Ethics Coordinator

The study follows a quantitative research methodology, as well as an experimental design that focusses on model performance evaluation through systematic validation techniques that include cross-validation and sensitivity analysis. The dataset samples will be publicly available game logs, as well as player logs from the past 3 seasons as well as the current season. This will be from both regular season and playoff games, allowing for balanced representation across varying competitive contexts.

The study will employ a variety of analytic tools, these include but aren't limited to:

- Python Data Preprocessing methods, for cleaning and normalisation, to then be used for feature engineering.
- Implementation of Pythagorean expectation from the data within the set itself.
- Application of supervised learning methods that can include logistic regression, decision tress, support vector machines and deep neural networks.
- Integration of the statistical insights from feature importance metrics and ensemble techniques.
- Model evaluation will be conducted with cross-validation (k-fold), sensitivity analysis, and comparison performance metrics such as F1-Scores and AUC, ROC and True/False Positive Rates (TPR & FPR).

A4 Will the project involve deceptive or covert research?	Yes / No
A5 If yes, give a rationale for the use of deceptive or covert research	
N/A	
A6 Will the project have security sensitive implications? NB: Any project which falls under this definition will be referred to the Prevent Co-ordinator by the Committee which is considering the application so it can be risk assessed in line with the Prevent Policy . Ethics approval will only be granted once the Prevent Co-ordinator is content with the outcomes of that risk assessment.	Yes / No
A7 If yes, please explain what they are and the measures that are proposed to address them	
N/A	

B PREVIOUS EXPERIENCE
B1 What previous experience of research involving human participants relevant to this project do you have?
N/A
B2 Student project only What previous experience of research involving human participants relevant to this project does your supervisor have?
N/A

C POTENTIAL RISKS
C1 What potential risks do you foresee? Include details of risks to the participants, the researcher and the project as a whole.
There are very few physical risks of the project, and with data being publicly available, the exploitation of private data is minimal to zero. However, to ensure the results of the study are reliable, the data and subsequent code used to analyse the data will need to be protected.
C2 How will you deal with the potential risks?
The data and the code will be stored in an account protected cloud facility, as well as only being accessed through the password protected PC that I always possess and keep on me. And to ensure that work will be continued if any files were to be corrupted or lost, a backup set will be saved to my removable hard drive.

When submitting your application you **MUST** attach a copy of the following:

- All information sheets
- Consent/assent form(s)
- Withdrawal of consent form

An exemplar information sheet, exemplar participant consent form and exemplar participant withdrawal form are available via the research section of the Cardiff Met website (see section on Ethics Governance). These are based on good practice and will be useful in the majority of cases. However, it is recognised that in some cases a project will be subject to requirements from an external body. Use of these exemplars is therefore not obligatory.

Appendix F – Github Repository Link

The link to find the raw code, pipelines and datasets used in this study is as follows:

<https://github.com/Strong649/Final-Project>

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